

Racial Retaliation and Updating: Black Athletic Leadership and Racial Hate

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Abstract

This paper studies how racial animus responds when minorities occupy highly visible leadership roles that are publicly praised and blamed for collective outcomes. We focus on Black quarterbacks in the National Football League (NFL)— a highly salient media environment central to national and regional identity in the United States. We show that exposure to Black quarterbacks shapes racial hostility in both the short and long-run. Using a game-day event-study design, we identify evidence of emotion-driven racial retaliation. Relative to games where both quarterbacks were white, losses involving a Black quarterback are followed by a 18% increase in hate crime rates. These effects are concentrated after emotionally engaging games which generate a 43% rise in hate crimes, a 29% increase in hate speech indicators, and a 0.043 sd increase in implicit bias. In the long-run, we document a performance-contingent updating process. Areas with low-performing Black quarterback-led teams exhibit little change in racial animus. In contrast, when Black quarterback-led teams advance to championship games, hate crime rates fall by 2.2%, racial slur searches decline by -0.029 sds, hate speech indicators decrease by 22%, and implicit bias measures fall by -0.001 sds. Exposure to opposition Black quarterbacks does not produce long-run effects.

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1 Introduction

Racial prejudice and racial hate continue to be pervasive problems in the United States. Reported incidents of anti-Black hate crimes increased by 84% between 2014 and 2023, while hate speech has proliferated on social media platforms. Prejudicial attitudes and behaviors continue to generate discrimination in education, health, employment, housing, and the criminal justice system (Alesina and La Ferrara (2014), Bayer et al. (2017), Lang et al. (2005)). In jobs and schools, in-group tolerance continues to favor majority group members in the allocation of blame and punishment (Egan et al. (2022), Sarsons (2017), Skiba et al. (2002)) particularly when minorities occupy positions of leadership, where they become focal points for judgments about competence, responsibility, and blame.

Recognition that racial biases and prejudices carry important social and economic costs has motivated a large literature aimed at understanding and reducing prejudice (Lang and Kahn-Lang Spitzer (2020), Rutland and Killen (2015), Paluck et al. (2021), Bertrand and Duflo (2017)). While media representations—and under-representation—are often cited as key mechanisms in the persistence of these biases, there is limited causal evidence on how exposure to minority public figures shapes attitudes. Sporting events provides a unique setting to fill this gap. They are highly salient, media-saturated events central to national and regional identity. Racially diverse teams compete under intense public scrutiny and emotion, and minority athletes often occupy highly visible leadership roles where they are evaluated, praised, and blamed. This setting allows us to study how majority groups respond when minorities occupy highly visible leadership roles that are publicly praised and blamed for collective outcomes. Do these responses depend on performance? And do they persist as individuals update their beliefs and social norms?

This paper studies these questions by examining how racial animus responds to minority leadership in team sports, focusing on Black quarterbacks in the National Football League (NFL). The NFL provides a particularly compelling context to study responses to visible minority leadership: teams are highly salient local institutions, and quarterbacks serve as highly visible leaders. The quarterback position is also historically racialized. Although Black athletes have long comprised the majority of players, Black quarterbacks remained rare for decades. As a result, Black quarterbacks have become especially salient figures and focal points for judgments about competence, authority, and responsibility for collective outcomes.. By examining responses to Black quarterbacks in NFL games, we provide new evidence on how leading players in one of America’s most visible cultural

institutions shape public attitudes and behavior toward African Americans, both around game days and over the course of a season.

First, using an event study design we investigate how racial animus changes in the very short-run – the five days surrounding games. We find evidence of a racial retaliation effect: racial animus increases when Black quarterbacks are perceived of as “blamable” for a negative outcome. By examining close scoring upset games that generate significant emotional responses, we find that relative to similar games where both quarterbacks were white, losses involving a Black quarterback are followed by a 43% increase in the anti-Black hate crime rate. In contrast, emotionally engaging wins involving Black quarterbacks leave hate crimes rates unchanged. This effect is observed on other measures of racial animus, including online hate speech indicators on reddit, which increases by 29%, and implicit racial bias which rise by 0.043 sds.

We also estimate the persistent effects of exposure to both local and opposition Black quarterbacks in the long-run. For local Black quarterbacks, we compare how outcomes evolve in areas with Black and white quarterback teams over the course of an NFL season. We find that Black quarterbacks on low performing teams have limited effects. In contrast, and consistent Mohamed Salah’s effect documented in Marble et al. (2021), sustained exposure to high-performing local Black quarterback teams leads to persistent reductions in racial hostility. In the months following the end of the NFL season, areas where Black-quarterback teams advance to the championships have hate crime rates that are 2.2% lower, Google searches for racial slurs decline by -0.029 sd, the share of reddit submissions moderated from team subreddits is 22% lower, and implicit racial biases drop by -0.001 sd. For opposition Black quarterbacks, we find no evidence that increased exposure to opposing Black quarterbacks – either through playing more games against Black quarterbacks or being knocked out of the playoffs by a Black-quarterback team – leads to persistent increases in racial animus.

These findings make several contributions. Most directly, this paper provides the first causal evidence that a central institution in American life—the NFL—can shape racial animus. Race has long been integral to the league’s social meaning. Black athletes have long played a prominent role in the league, and the quarterback position has remained a focal point for debates over leadership and evaluation. The NFL has therefore both reflected and shaped broader American debates over race, merit, authority, and belonging. We show that NFL games do not merely generate generic emotional responses. They can trigger short-run racial retaliation when minority leaders

are associated with failure, and they can generate more persistent reductions in racial animus when local minority leadership is repeatedly associated with success. In doing so, the paper shows that one of America's most visible nonpolitical institutions can shape social cohesion within the United States.

The paper also contributes to the literature on the effects of minority leaders and mass media on prejudice. A growing body of work shows that exposure to hostile messages transmitted through film, radio, and social media can worsen inter-group bias, hate, and persecution (DellaVigna and La Ferrara (2015), Yanagizawa-Drott (2014), Adena et al. (2015), Ang (2023), Müller and Schwarz (2021), Müller and Schwarz (2023), Cao et al. (2023)). Much less is known about whether media-based exposure to minority public figures can reduce prejudice, or when such exposure instead generates backlash. This gap is especially important in settings where minority figures are not only visible, but occupy leadership roles and are repeatedly praised and blamed for collective outcomes. While a substantial literature in economics has examined minority representation in leadership and role-model positions, this work has focused primarily on academic and aspirational outcomes for underrepresented groups (Riley (2024), Gershenson et al. (2022), Lim and Meer (2020), Fairlie et al. (2014), Dee (2005)). Research on majority-group responses to minority leadership has instead focused mainly on political settings (Sakong (2026), Beaman et al. (2009)), where reactions may partly reflect policy preferences and ideological conflict rather than minority leadership itself. The NFL provides a context in which leaders are repeatedly evaluated on observable performance and publicly praised and blamed for collective outcomes, features that closely parallel many leadership environments outside politics. The paper therefore speaks to a central and less understood question: whether minorities are accepted not only as participants, but as leaders, when their performance is repeatedly observed and evaluated.

Evidence of prejudice reduction due to media based contact with apolitical out-group role models is fairly limited (Paluck (2009), Blouin and Mukand (2019), Duquenois and Zeng (2025), Plant et al. (2009), Miller et al. (2020)). Particularly relevant is Marble et al. (2021), who documents a decline in anti-Muslim hate crime and hate speech in Liverpool following the arrival of the highly successful soccer star Mohamed Salah in 2017. Our findings are consistent with the effects they document, but make several important advances. First, we move beyond the study of a single famous successful individual. By comprehensively examining the impact of 68 Black quarterbacks over 23 NFL seasons we establish that their findings are a generalizable reaction to minority athletes in leadership positions rather than an idiosyncratic individual effect. Second, we examine the

effect of playing against opposition minority players, allowing for a more complete picture of how minority athletes affect these outcomes not just in their team's region but in opposition teams' regions as well. Third, we examine how these impacts crucially depend on performance, finding a pattern of racial retaliation. Finally, we examine both the effects of persistent long-run exposure to minority players, as well as the emotion driven responses that happen right around games using multiple measures of racial animus that span both concrete criminal behavior (anti-Black hate crimes) and more latent, attitudinal expressions (online hate speech, racial slur internet searches, and implicit racial biases).

Finally, this paper contributes to the literature on the economic and social impacts of sporting events. Research has shown that sporting events generate powerful emotional shocks with important social consequences, including effects on domestic violence (Card and Dahl (2011), Matsuzawa and Arnesen (2024)), other violent crimes (Rees and Schnepel (2009), Munyo and Rossi (2013), Montolio and Planells-Struse (2016), Ivandić et al. (2024)), judicial decisions (Eren and Mocan (2018)), financial markets (Edmans et al. (2007)), fertility (Stoecker et al. (2016)), and infant health (Duncan et al. (2017)). Our short-run results show that these emotional shocks can be racialized: games can generate retaliatory behavior toward out-groups who share an identity with salient athletes. A couple of existing papers have examined the impact of international sporting tournaments on national identity building. Depetris-Chauvin et al. (2020) show that international soccer tournaments strengthen national identities and reduce civil conflict, and Rosenzweig and Zhou (2021) document a similar effect, with a concurrent increase in negative views towards foreign refugees. Our work contributes to our understanding of the impacts of sports on co-national identities by showing that sports can impact attitudes towards minority groups, and thus social cohesion between groups within a country. Furthermore, most of this literature has focused on effects resulting from emotional shocks that occur right around game days. Our season level specification allows us to examine the longer-run effects of repeated exposure to a Black quarterback revealing sustained changes in behaviors.

The remainder of this paper is organized as follows. Section 2 presents contextual information on the NFL and Black quarterbacks in the NFL. Section 3 outlines a simple model. Section 4 presents our data. Section 5 focuses on our short-run analysis, presenting first our estimation strategy followed by our short-run results. Section 6 develops our season analysis beginning with the estimation strategy and then presenting results. Section 10 concludes.

2 The NFL and Black quarterbacks

The National Football League (NFL) is the highest level of professional American football and the most popular sporting league in the United States, receiving high levels of media attention. It is estimated that Super Bowl games, which determine the season's winning team, are watched by over 40% of the US population. Yet even regular season games are highly visible events averaging over 16 million viewers. The league is composed of 32 teams. The regular season, which typically runs from September to January, features weekly games. These often occur on Sundays (85% of games) though teams occasionally play on Mondays (7%), Thursdays (5%), and Saturdays (4%). The regular season consisted of 16 games until the 2021 season when it was increased to 17 games per season. During the regular season, games are determined by a scheduling formula. After the regular season, the best teams qualify for the playoffs (12 teams prior to the 2020 season and 14 teams thereafter). The playoffs, typically running throughout January, are a knockout tournament that leads to the "semi-final" conference championship games (4 teams), and finally to the Super Bowl, usually held in the first two weeks of February.

Games are highly structured, with frequent stops in play, highly specialized positions, and players who take the field only for offense or defense. Quarterbacks lead the team's offense and must make quick decisions under pressure—they receive the ball at the start of most plays and implement the team's play strategy. As a result, quarterbacks are often a team's most visible and influential player, receiving a substantial share of media attention, and the position is a symbol of leadership and masculinity in American culture. Though the NFL began reintegrating Black players in 1946, for decades Black athletes were largely excluded from the quarterback position due to racist stereotypes questioning their intelligence, leadership, and decision-making ability (Rhoden (2007), Reid (2022)). In 1953, Willie Thrower broke that barrier as the first Black quarterback to play in an NFL game, and in 1968, Marlin Briscoe became the first to start as quarterback in the modern era. Though Black athletes have long comprised the majority of NFL players, roughly 70% since 1990, the number of Black quarterbacks rose only gradually from the late 1970s onward (Marquez-Velarde et al. (2023)).¹ In our period of study, about 20% of quarterbacks were Black

¹Racial discrimination within professional sports has been studied fairly extensively. We know that racial biases have influenced referee and umpire calls (Price and Wolfers (2010), Parsons et al. (2011), Gallo et al. (2013)) and that the racist behavior of fans impacts player performance (Caselli et al. (2023), Caselli et al. (2024)). In the NFL, several studies have explored the effects of player race on careers and pay with mixed findings, reflecting the wide variety of seasons, positions, and approaches employed for analysis (most recently Berri et al. (2023), Gregory-Smith et al. (2023), Keefer and Kniesner (2023), Keefer (2013), Burnett and Van Scyoc (2015), Ducking et al. (2014) Ducking et al. (2017)). With respect to

between 2002 and 2020 with a notable increase afterwards to reach 37% in 2024 (Figure 1). Almost all NFL players are American.

3 A model of Black athletes' impacts on racial animus

We generate a parsimonious model of how exposure to NFL Black quarterbacks impacts racial animus in a team's catchment area, focusing on the two mechanisms most consistent with our findings: emotion-driven racial retaliation in the short run and performance-dependent parasocial updating in the long run.

Let areas contain a continuum of individuals, i , who, after their team n 's game g in season s , choose to act upon their racial animus or not. The additional utility i receives from taking such an action after game g is

$$U_{ing} = \theta_{ins} + r_{ng} - c_{ins} + \varepsilon_{ig}, \quad (1)$$

where θ_{ins} captures i 's baseline racial animus, r_{ng} captures the short-run differences in racial animus generated by game g for team n , c_{ins} is an individual-specific cost to the action, and ε_{ig} is an i.i.d. shock. In the short-run, changes in racial animus right around game day reflect changes in r_{ng} , as baseline racial animus and cost do not vary in the short-run. Longer run effects of season exposure to Black quarterbacks are captured by changes in θ_{ins} .

Short-run emotion driven racial retaliation (r_{ng}): We use the term racial retaliation to describe a process whereby members of a majority group direct racial animus toward innocent out-group bystanders when a salient out-group member is perceived as "blamable" for a negative outcome. Experimental evidence shows that, following negative outcomes, majority group members are more likely to attribute responsibility to minority individuals—even when those individuals are not causally responsible (Bursztyn et al. 2022; Bauer et al. 2023), a pattern often referred to as *racial scapegoating*. Racial retaliation differs from racial scapegoating in that it requires the negative outcome to be associated with a racially salient out-group figure who can plausibly be blamed. The hostility is then displaced onto the broader group.

We model games as generating affective responses – increasing in wins and decreasing in losses–

quarterbacks, it is widely acknowledged that the position was historically much less accessible to Black players. In more recent periods, some evidence suggests Black quarterbacks experience differential treatment in benching and pay (Volz (2017), Berri and Simmons (2009), Berri et al. (2023)).

that are more pronounced for emotionally engaging games such as close games with unexpected outcomes. The affect generated by game g for team n is

$$a_{ng} = (\gamma_0 + \gamma_1 Engaging_{ng}) \cdot (\alpha_1 Win_{ng} + \alpha_2 Loss_{ng}). \quad (2)$$

Affect shocks generate retaliatory racial animus when they are attributed to racially salient agents in the game like Black quarterbacks. We model this blame mapping as

$$r_{ng} = a_{ng} \cdot (\pi_l l_{ng} + \pi_o o_{ng}), \quad (3)$$

where $\pi_l, \pi_o \geq 0$ capture whether changes in affect are attributed to Black quarterbacks and generate retaliatory black animus towards local Black populations. Note that π_l and π_o allow for the magnitude of this effect to depend on whether the Black quarterback was playing for the local or opposition team.

Long-run performance dependent updating (θ_{ins}): Repeated exposure to Black quarterbacks – who occupy a highly visible and admired leadership position – may shift racial attitudes by altering beliefs about competence, leadership and social norms. Over time, such exposure might generate stereotype revision, norm updating, or changes in priors about group characteristics.

The parasocial contact hypothesis posits that repeated media exposure to positively portrayed out-group members can reduce prejudice (Schiappa et al. (2005)). While a generalized version of this hypothesis predicts declines in animus from repeated exposure to Black quarterbacks regardless of performance, a performance-mediated version allows for more nuanced, outcome-contingent effects: positive performance—especially by local quarterbacks—should produce larger reductions in animus by providing favorable signals. Evidence that anti-Muslim hate declined in Liverpool after the highly successful Mohamed Salah joined Liverpool FC (Marble et al. (2021)) is consistent with this mechanism. In contrast, poor performance may dampen positive updating or even reinforce negative priors, particularly in emotionally charged, media-saturated environments that amplify blame. Poor performances by minority athletes could therefore generate persistent racial backlash—a hypothesis that has not yet been empirically tested.

With these mechanisms in mind, the effects of exposure to opposition quarterbacks are theoretically ambiguous. Such exposure is less frequent and occurs in a markedly different context than exposure to local players. Although opposition minority athletes may increase the salience

of minority leadership, media portrayals are less likely to be positive, and the competitive setting lacks a shared local identity. This context may heighten social boundaries and reinforce group distinctions rather than attenuate them. Consequently, the long-run effects of opposition exposure depend on how performance signals interact with identity and attribution processes—an empirical question that remains unexamined.

Accordingly, we flexibly model baseline racial animus, θ_{ins} , as a slow-moving state variable that evolves through repeated exposure to Black quarterbacks and their team’s performance, with effects that can differ for local and opposing Black quarterbacks. Attitudes update as individuals partially adjust their existing attitudes toward a season-specific signal observed for team n during season s . The change over the course of the season is given by

$$\theta_{in,s+1} - \theta_{ins} = \lambda_L \left(\tilde{\theta}_{ns}^L - \theta_{ins} \right) + \lambda_O \left(\tilde{\theta}_{ns}^O - \theta_{ins} \right) + \varepsilon_{ins}, \quad (4)$$

with

$$\tilde{\theta}_{ns}^L(L_{ns}, Rec_{ns}), \quad \tilde{\theta}_{ns}^O(O_{ns}, \Phi_{ns}), \quad \lambda_L, \lambda_O \in (0, 1). \quad (5)$$

λ_L and λ_O govern the rate of updating while $\tilde{\theta}_{ns}^L$ and $\tilde{\theta}_{ns}^O$ summarize the cumulative effect of exposure to local and opposition Black quarterbacks during the season. The first exposure measure, $\tilde{\theta}_{ns}^L$, is a function of exposure to a local Black quarterback, L_{ns} , and the team’s performance record, Rec_{ns} . The second exposure measure, $\tilde{\theta}_{ns}^O$, captures exposure to Black quarterbacks on opposing teams, O_{ns} , and the outcome of these interactions, Φ_{ns} . ε_{ins} captures idiosyncratic shocks to animus unrelated to football exposure.

4 Data

4.1 Measuring exposure to Black quarterbacks

To measure exposure to Black quarterbacks, we proceed in two steps. First, we delineate each NFL team’s geographic catchment area. Second, using play-by-play game data, we identify games and seasons where a Black quarterback served as the team’s primary quarterback.

Defining NFL teams’ geographic catchment areas: We match U.S. Designated Market Areas (DMAs), which delimitate local media markets, to NFL teams by estimating DMAs’ ratings for

each of the 32 NFL teams. For each DMA-team combination we calculate

$$\widehat{Ratings}_{DMA}^{team} = HomeRatings^{team} \times GoogleSearches_{DMA}^{team}.$$

$HomeRatings^{team}$ is an estimate of a team's ratings in the team's local DMA.² $SearchIndex_{DMA}^{team}$ is built using Google trends indexes of DMA searches for each team over the 2004-2025 time period.³ DMAs are then matched to their highest rated team. NFL team catchment areas thus defined are mapped in Figure 2.

NFL data: Using public APIs, we collect play-by-play NFL data for every game in the 2002 thru 2024 seasons. In this data we observe every action players took on the field during the game. Though only one quarterback can be on the field at once, teams can field multiple quarterbacks over the course of a game if their starting (main) quarterback is injured or is otherwise removed from play. We define a team-game's lead quarterback as the quarterback with the greatest share of plays during the game. The race of the quarterbacks is then identified by a team of research assistants. For our season level analysis, we define team-seasons as having a Black quarterback if the majority of the team's regular season games are defined as having a Black quarterback.

Overall, this data covers 6214 games and 736 season-teams.⁴ Of these, we omit from our analysis 345 games and 23 season-teams where the main quarterbacks is one of the two non-Black Hispanics or two Samoan quarterbacks in our data, giving us a sample composed of European-American and African-American quarterbacks. For our short run estimations, which examine outcomes in 5 day windows centered around game days, we further restrict our analysis to team-games that occur at least 7 days after the team's previous game and 3 days before the subsequent game. Finally, for brevity and clarity of exposition, our game-level analysis also omits 307 games that

²Disaggregated ratings data is not easily available. To estimate this we take the average of the team's ratings in teams' local media markets as reported for five available seasons (2011, 2012, 2013, 2020, 2021) in the Sports Business Journal (Karp (2014); Ourand (2022)).

³The Google trends indexes we use are requisitions of DMA level Google trends indexes for each team over the 2004-2025 time period. These are set to 1 for the DMA with the greatest interest in the team over the 2004-2025 period with the index for other DMAs interpretable as searches relative to the top searching DMA. Google trends requisitions are calculated using a sample of Google searches (rather than the universe of Google searches that were conducted during a period), with the sample changing on a regular basis. To ensure that we generate a consistent search index measure, we collect 8 separate requisitions and calculate a team's search index measure as the mean index across these 8 requisitions. To verify that our measure is stable, we correlate this with the average calculated with only the first 7 requisitions finding a correlation of 1.00.

⁴Each of the 32 teams played 16 regular season games between 2002 and 2020, and 17 between 2021 and 2024. Playoffs consisted of 11 games between 2002 and 2019 and 13 games between 2020 and 2024. We drop the 2022 week 17 game between the Buffalo Bills and the Cincinnati Bengals that was canceled early in the first quarter due to Bills player Damar Hamlin's medical emergency.

feature two Black quarterbacks as these are few in number and confound multiple effects, complicating interpretation. With these restrictions, our final game-level data covers 5562 games. Of these, 3501 are classified as having two white quarterbacks, while 2061 have a Black quarterback on the field. At the season level, 151 of 713 season-teams are classified as having a Black quarterback.

4.2 Measures of racial attitudes and hate

Anti-Black hate crimes: We use the FBI’s Uniform Crime Reporting Hate Crime Statistics for 2002–2025.⁵ These data are the most comprehensive standardized police-reported measure of hate-crime incidents available in the United States and are widely used in hate-crime research (Müller and Schwarz 2023; Anderson et al. 2020; Mulholland 2013). Nevertheless, it is important to note that reported incidents documented in these data are only a small share of all hate crimes. Many hate crimes are never reported to law enforcement, and many local law enforcement agencies do not reliably report hate crimes to the FBI as reporting is voluntary (Dharmapala and Huq (2024)).⁶ The incident-level files report the calendar date and location of each incident, a bias-motivation category identifying the targeted group, and the reporting agency. We map incidents to counties using the locations of over 24,000 reporting agencies. We then calculate the count of anti-Black incidents on each day for each DMA to generate the daily DMA incident rate (per 10 million). For our season level analysis, day counts are aggregated to generate monthly DMA incident rates (per 100,000). Throughout, we check that our main findings are robustness to alternative measurement and specifications of hate crimes.⁷ After matching this data with NFL games, our final sample spans the 2002 to 2024 seasons covering 10257 unique games-teams and 682 unique season-teams.

Reddit data: We collect reddit submissions from 2010 to 2024.⁸ Our sample is composed of 32 subreddits dedicated to discussion of the 32 NFL teams. For each subreddit-day we calculate the

⁵In this data, a hate crime is defined as a criminal offense motivated, in whole or in part, by the offender’s bias against a protected characteristic such as race.

⁶Between 2011 and 2015, 54 percent of hate crimes identified in the National Crime Victimization Survey were not reported to the police (Dharmapala and Huq (2024)).

⁷These include an extensive indicator specification, a zero adjusted log specification, linear and Poisson specifications on incident counts and the inclusion of population weights.

⁸We use reddit data from 2010 onward for the following reasons. First, prior to 2010, reddit’s overall activity was low and many subreddits either did not exist or were nascent, which would induce severe left-truncation and compositional shifts across units. Second, HATEBERTA was developed and tuned on 2010s social-media text; beginning in 2010 mitigates distribution shift between the training corpora and our corpus.

daily count of submissions to the subreddit.⁹ We then calculate the share of submissions that were removed by content moderators as our first indicator of hate speech.¹⁰ For the submissions with visible content, that were not removed by moderators or deleted by authors, we generate a measure of anti-Black hate speech. We use **HATEBERTA**, a transformer-based classifier trained for abusive-content detection on reddit (Caselli et al. 2020).¹¹ To obtain a measure of anti-Black targeted hate-speech, we further fine-tune **HATEBERTA** using **ToxiGen** (Hartvigsen et al. (2022)) to generate a 0 to 1 score interpretable as the probability the text contains anti-Black targeted hate-speech. Using this measure, we find minimal amounts of anti-Black hate speech in the submissions to official NFL team subreddits that were not removed by moderators. The ToxiGen estimated probability that a submission contains anti-Black hate is 0.019% on average and only 0.04 % of submissions have an estimated probability greater than 0. Accordingly, we treat the moderation share as our primary indicator of hate speech activity. Estimates using the ToxiGen score are imprecise and reported in the appendix. After matching the reddit data with NFL games, our final sample spans the 2009 to 2024 seasons, covering 4320 unique game-teams and 244 season-teams.

Racial Animosity Index (RAI) using Google search indices: We construct a location and time varying measure of anti-Black racial animus using the index of Google searches for racial slur queries.¹² This measure, first developed by Stephens-Davidowitz (2014), has been widely used in the literature as a measure of racial animosity (Chetty et al. (2020), Kline et al. (2022), Derenoncourt (2022)). The slur query is most frequently searched for along side the term “joke(s)” which returns offensive websites. Searches that include the slur query are not uncommon. From 2004-2025, such searches are about as common as searches for the terms “cavity+cavities”.

To capture geographic and temporal variation in slur searches, while still maintaining suffi-

⁹We focus on Reddit submissions rather than comments for several reasons. First and foremost, comments associated with submissions that were moderated off of the platform are unobserved resulting in a selected sample at the comment level. Second, submissions are more likely to reflect deliberate and sustained expression, whereas comments frequently capture rapid reactions and conversational back-and-forth, often consisting of exclamations or fragmentary expressions with limited semantic information. This substantially increases measurement error when estimating hate-speech intensity using text-based classifiers.

¹⁰Reddit historically relied on decentralized, volunteer-based moderation and imposed few platform-wide restrictions prior to 2015. From 2015, reddit began banning prominent subreddits associated with harassment and hateful content, eventually introducing an explicit platform wide hate-speech policy in 2020 (Reddit 2020; Chandrasekharan et al. 2017). In addition to these platform-wide policies, NFL team subreddits enforce their own community-specific content rules, which commonly prohibit hate speech and harassment, as well as other nuisance content such as spam, off-topic discussions, and cross-posting.

¹¹**HATEBERTA** is trained on a broad spectrum of hate speech categories, including those targeting racial and ethnic groups (e.g., Black, Asian, Latino).

¹²Specifically, we requisition indexes for searches that include the terms “n___ +n___s” — henceforth “the slur query.”

cient search volumes for index construction, we construct the standardized index for DMAs in two-month time bins using the mean index calculated across ten requisitions from Google trends.¹³ After matching this index with NFL seasons, our final sample spans the 2004 to 2023 seasons, covering 613 team-seasons.

Implicit association test (IAT) scores: Measures of implicit biases come from the publicly available data on implicit association tests collected online by *Project Implicit*. *Project Implicit* allows interested parties to take a short survey and implicit association tests (IATs) generating a measure of their implicit associations with individual characteristics such as race. IATs are widely used in psychology, and increasingly in economics, as a way of measuring implicit attitudes (Glover et al. (2017), Carlana (2019), Corno et al. (2022), Lowes et al. (2015)).¹⁴ After matching the IAT survey data with NFL games, our final sample spans the seasons from 2002 to 2024, covering 6019 unique game-teams and 618 season-teams.

Table 1 provides descriptive statistics for our main outcomes measured at both the day (panel (a)) and season (panel (b)) levels.

5 Game-day Responses: Racial Retaliation

In this section, we begin by examining the impacts of exposure to Black quarterbacks on and after game days. Previous work has shown the importance of powerful emotional responses to game outcomes in the days that follow games (Card and Dahl 2011; Montolio and Planells-Struse 2016;

¹³For each requisition, the Google index is set to 1 for the DMA with the greatest share of slur queries while the index for other DMAs are interpretable as search intensity relative to the top searching DMA. Google trends requisitions are calculated using a sample of google searches (rather than the universe of Google searches that were conducted during a period), with the sample changing on a regular basis. To ensure that we generate a consistent measure, we collect ten requisitions for every two-month bin and calculate the mean across these ten requisitions. As Google does not return an index when samples are too small, we drop any mean calculated with fewer than two observed requisitions (25.4% of DMA-2month bins, mostly in the earliest years of the data and low population DMAs). We then calculate the mean index across these ten requisitions which is then standardized.

¹⁴When completing the race IAT, respondents are sequentially presented with images of Black and white individuals, and words that have positive or negative connotations. Respondents complete a series of rapid sorting exercises grouping together the images with words. The IAT is meant to measure the strength of a respondent's association between individual characteristics and word connotations, as sorting is easier when associated items are sorted together. IAT measures have been the subject of increased scrutiny in the psychology literature (Ratliff and Smith (2021)). There is generally agreement that these measures are relevant and predictive, particularly for socially sensitive topics (Greenwald et al. (2009), Bertrand and Duflo (2017)), and aggregate regional measures (Hehman et al. (2019)). Disagreements have centered on the validity of the *implicit* bias construct, if it differs from *explicit* biases, and whether the low test-retest reliability is due to high measurement error, or the construct itself being time variant (Gawronski (2019), Schimmack (2021), Connor and Evers (2020)).

Ivandić et al. 2024; Matsuzawa and Arnesen 2024). We find that beyond simple emotional reactions to wins and losses, we observe evidence of a racial retaliation effect. Racial animus increases after lost games when the game featured a Black quarterback, regardless of whether they were playing for the local or opposition team. This racial retaliation effect is driven by engaging emotional close-upset games, consistent with findings in Card and Dahl (2011) and Matsuzawa and Arnesen (2024) that show how such games fuel emotion driven responses. Below we explain our game day empirical strategy and present these findings.

5.1 Game-day empirical strategy

Our game-day design estimates how outcomes in the days that immediately follow a game change based on the racial composition of the quarterbacks that played, the game’s result, and expectations. Using the NFL game data and the data collected on player race, we classify game g for NFL team n as having a Black quarterback ($QBBlack_{ng} = 1$) if a Black player played that position for the majority of the games’ plays. For each game-team we then define C , the game’s quarterback composition (QBC_{gn}) from each team’s perspective. For our main specification, we examine two different quarterback compositions: $w = 1$ games where both quarterbacks are white (eg. white-white games) and $b = 1$ games where any quarterback (opposition or local) is Black. Heterogeneity between these is examined subsequently. For each of these compositions, we examine how outcomes change in five day windows centered around game days ($d = 0$). We use a pre-post specification where $Post_{dg}$ is set to 1 for the game day and the two subsequent days, and 0 for the two days prior. $Post_{dg}$ is interacted with our indicators for the game’s quarterback composition and whether the game was won ($Win_{gn} \in \{0, 1\}$).¹⁵ We estimate

$$\begin{aligned} Y_{agnd} = & \rho_0 + \rho_1^w Post_{dg} + \rho_2^b Post_{dg} \times \mathbb{1}\{QBC_{gn} = b\} \\ & + \rho_3^w Post_{dg} \times Win_{gn} + \rho_4^b Post_{dg} \times \mathbb{1}\{QBC_{gn} = b\} \times Win_{gn} \\ & + \eta_{ag} + \omega_d + \epsilon_i, \end{aligned} \quad (6)$$

where Y_{agnd} is an outcome for area a , team n , on day d , around game g . A day of the week fixed effect, ω_d , controls for weekly trends in outcomes. Importantly, we include a $area \times game$ fixed effect, η_{ag} , which restricts our estimating variation to outcomes measured around the same game for the same geographical unit (and thus the same team). This fixed effect absorbs baseline racial

¹⁵Ties are uncommon in the NFL during this period. In our game data, only 14 of 6214 games resulted in a tie. These are coded as $Win_{gn} = 0$.

animus, as well as any cost associated with the expression of racial animus, which we treat as fixed in the short-run.

ρ_2^b in equation 6 identifies how the change in outcomes after games differs for lost games involving a Black quarterback, as compared to white-white quarterback losses. $(\rho_2^b + \rho_4^b)$ gives this differential for wins. As racial retaliation is a response to the affect shock generated by the game, we estimate these effects separately for emotionally engaging and unengaging games. We use close-upsets games to identify emotionally engaging games. These are close scoring games where the final score differential is 7 or less (i.e. one touchdown determined the games outcome), that are also upset games in that the pre-game spread predicted that the losing team would win by 3 points or more. Unengaging games are those that are neither close scoring nor upsets.

For figures we estimate equivalent event time specifications where coefficients are calculated for each day relative to the day before the game.

5.2 Short-run effects of Black quarterbacks

Anti-Black hate crimes: Table 2, rows 3 and 4, reports our main short-run estimates for local Black quarterback games on DMAs' daily hate crime incident rates (per 10 million). We find that hate crimes increase after losses involving a Black quarterback due to these games activating an emotion driven racial retaliation effect.

Column 1 reports estimates of equation 6 on all games in our analysis sample. On average, we find that the hate crime rate differentially increases by 0.037 (p-value=0.005), a 18% increase, after losses involving a Black quarterback as compared to the change observed after losses where both quarterbacks were white (which do not significantly impact hate crimes – per row 1). Following wins, the differential increase in the hate crime rate, $(\rho_2^b + \rho_4^b)$, is much smaller at 0.015, and we cannot reject the null of no effect (p-value=0.22).

This pattern is driven by emotionally engaging games. We follow the literature and examine the effects around games that generate significant emotional engagement and shock. Column 2 estimates equation 6 on close-upset games finding much larger effects. After a close-upset loss involving a Black quarterback, the hate crime rate differentially increases by 0.086 (p-value=0.008), a 43% increase. In contrast, the $(\rho_2^b + \rho_4^b)$ point estimate of -0.018 suggests a reduction in the hate crime rate following wins involving Black quarterbacks, though this is not statistically significant (p-value=0.60). The first two panels of Figure 3 present event day estimates for these close-upset games. For reference, panel (b) plots event day estimates for white-white emotionally engaging

losses (ρ_1^w) and emotionally engaging wins ($\rho_1^w + \rho_3^w$). Panel (a) plots the differential effects for emotionally engaging losses (ρ_2^b) and wins ($\rho_2^b + \rho_4^b$) involving Black quarterbacks. Consistent with the aggregated estimates reported in the table, panel (b) shows no evidence of any impact around white-white games while panel (a) shows large differential effects that begin on game day and persist for the next two days.

The emotional engagement these games create is key to generating these effects. In columns 3 and 4 we split the sample between games that may generate some level of emotional engagement, having either a close score and/or an upset outcome (column 3), and unengaging games that are neither close scoring nor upsets (column 4). Though the point estimate for losses in unengaging games is positive, it is small and not statistically significant at 0.019 (p-value=0.34). These patterns of large increases after emotionally engaging losses involving Black quarterbacks with little to no change following other games, be they wins or unengaging games, are robust to alternative specifications of our measure of hate crimes and alternative combinations of fixed effects (see appendix tables A1 and A2). They are also observed in our other measures of racial animus.

In appendix Table A3 we examine impacts on hate crimes targeting white, Hispanic and other racial groups. Consistent with a retaliation mechanism, we find no evidence of game-day changes in hate crimes targeting White or other racial groups. Hate crimes targeting Hispanics are less common. They do exhibit patterns similar to those directed at Black populations, although the effects are somewhat smaller in magnitude. It is worth noting that Hispanics encompass a racially heterogeneous population, including individuals with darker skin tones, some of whom also identify as Black.

Hate speech and IAT scores: Table 3 repeats the same analysis on the two other measures of racial animus that are observable at the day level: the share of submissions moderated off of NFL team subreddits, and white IAT scores. For both these outcomes, we see strikingly similar patterns: differential increases following close-upset losses involving Black quarterbacks (row 3 of columns 2 and 6), and little to no differential effect following wins, or after unengaging games.

For submission removals, we find that the share of submissions being moderated off of reddit experiences a general increase after all games (rows 1 and 2). This increase is substantially greater following emotional losses involving Black quarterbacks. The share of submissions being removed differentially increases by 0.045 (p-value=0.010), a 29% increase, after emotional losses involving a Black quarterback as compared to the change observed after such losses where both quarterbacks

were white. Event day estimates for these close-upset games are illustrated in the second two panels of Figure 3. Panel (d) shows the general increase in moderation share around game days while panel (c) shows the large differential increase in the days following losses involving Black quarterbacks. We see no evidence of a differential change following engaging wins (0.006; p-value=0.65) or around unengaging games (column 4). Appendix Table A4 and Figure A1 examine whether the number of submissions is differentially affected by quarterback race. As expected, submission activity on team subreddits increases after games but we find no evidence of a differential effect around Black quarterback games.

Estimates on white respondents' IAT test scores are slightly underpowered but show a similar pattern. IAT scores differentially increase by 0.043 sds (p-value=0.06) after emotionally engaging losses involving a Black quarterback. We see no statistically significant evidence of a differential change following engaging wins (-0.031 sds; p-value=0.21) or around unengaging games (column 8). Event day estimates for close-upset games are illustrated in the last two panels of Figure 3. As anyone can log onto the *Project Implicit* website and take an IAT test, in appendix Table A4 and Figure A1 we examine whether the number of test takers around these games is differentially affected by quarterback race. We see no evidence that the number of IAT test takers changes around game day.

The asymmetric response to Black quarterbacks – heightened animus after emotionally engaging losses but not after wins – is most consistent with these games triggering a short-run racial retaliation mechanism. To further confirm this, we estimate equation 6 disaggregating our analysis to separately examine the effects of local ($C = l$) and opposition ($C = o$) Black quarterbacks. Estimates are reported in Table 4. Across all three outcomes we observe similar patterns. Racial animus indicators increase following engaging losses but not after engaging wins involving Black quarterbacks, regardless of whether the Black quarterback is playing for the local or opposition team. Though there are some differences in the point estimate magnitudes for local (ρ_2^l) and opposition (ρ_2^o) Black quarterbacks, all the ρ_2 estimates are positive. For all three outcomes, we fail to reject that ρ_2^l and ρ_2^o are equal. Similarly, all the corresponding ρ_4 estimates are negative, suggesting offsetting effects following wins. For all three outcomes, we fail to reject that $\rho_2^l + \rho_4^l = \rho_2^o + \rho_4^o$. Overall, there is no clear evidence of effect heterogeneity based on whether the Black quarterback is playing for the local or opposition team. This is consistent with a racial retaliation effect where either the local or opposition Black quarterback is “blamable” for the the negative affect shock.

While our findings provide evidence of short-run emotion-driven racial retaliation, other mechanisms may also operate in this setting. For example, the presence of Black quarterbacks in historically white, high-status positions could trigger a status threat mechanism, whereby members of dominant groups respond with increased out-group animosity when their relative position is perceived as challenged (Blumer (1958), Blalock et al. (1967)). Status threat would predict increases in animus following exposure to Black quarterbacks, particularly after wins over white quarterbacks. However, this pattern does not align with our estimates in Table 4 showing that animus rises following losses by local Black quarterbacks – a pattern consistent with affective retaliation but not status-based backlash. More broadly, multiple mechanisms may act simultaneously. While we cannot definitively rule out alternative processes, the empirical patterns we observe are most consistent with a pattern of short-run emotion-driven racial retaliation. Other mechanisms may contribute at the margin, but this mechanism is necessary to account for the observed results.

6 Season responses: Performance sensitive updating

The patterns described in section 5 show a racialized emotional response to the performance of Black quarterbacks by comparing outcomes for the same location in the days right around game days through the use of location-game fixed effects. While effective at isolating immediate short run responses, the use of these fixed effects absorbs any long term effects that repeated exposure to Black quarterbacks could generate. Marble et al. (2021) show evidence of reduced anti-Muslim attitudes for a year following Mohamed Salah’s move to Liverpool in 2017. In this section we build on these finding in three important ways. First, we examine whether this long term pattern is generalizable, estimating it for a different sport and examining a large group of Black quarterbacks, helping to alleviate concerns about idiosyncratic effects tied to a single individual. Second, we examine how this pattern responds to performance. And third, we examine whether fan bases of teams that confront these players as opponents display any evidence of changed behavior as a result in the long-run.

Consistent with the findings in Marble et al. (2021), we observe reductions in anti-Black attitudes and hate in locations with Black quarterbacks on high performing NFL teams that make it to the

conference championships (4 teams per season). In contrast, Black quarterbacks on low performing teams have limited effects. With respect to opposition Black quarterbacks, we find no evidence that playing against Black quarterbacks has any long-run effects on racial animus. Below we explain our estimating strategies and present these results.

6.1 Local Black quarterbacks in the long-run

6.1.1 Empirical strategy

Our long-run design estimates how the evolution of outcomes over the NFL season differs based on quarterback race and team performance. For the time period under consideration, the NFL regular season begins in early September and ends in early January. This is followed by playoff games in January, culminating in the Super Bowl which generally occurs in the first or second week of February. We define seasons as running from May to April so as to include measures of our outcomes both before and after NFL games are taking place. For tables, we divide seasons into three time periods T such that $T \in \{b, d, a\}$. We set our omitted period to b , the period before the season start (May thru August) which we compare to d , outcomes observed during the NFL season (September thru February); and a the after season (March and April) – our period of interest to observe persistent impacts. For figures, to better observe the evolution of outcomes over the course of the season, we divide seasons into six two-month bins, with $T \in \{mj, ja, so, nd, jf, ma\}$, omitting the July-August bin. We define a team n in season s as having a Black quarterback ($L_{ns} = 1$) if the majority of the team's regular season games are classified as being played by a Black quarterback. Finally, a team's season record (Rec_{ns}) is categorized into three levels R such that $R_{ns} \in \{r, p, c\}$. These are: r , our omitted category, for the 450 team-seasons who only played in the regular season, p for the 194 team-seasons that advance to the playoffs but not the championships, and c for the 92 team-seasons that advance to the championships. Our main specification estimates the differential effect of having a Black quarterback as compared to a white quarterback for teams with different season records as follows:

$$Y_{ansm} = \mu_0 + \sum_T \mu_T^r \mathbb{1}\{m = T\} \times L_{ns} + \sum_T \sum_{R \neq r} \mu_T^R \mathbb{1}\{m = T\} \times L_{ns} \times \mathbb{1}\{Rec_{ns} = R\} \\ + \tau_{as} + \kappa_{am} + \psi_{Rm} + \phi_{ms} + \epsilon_i. \quad (7)$$

Y_{ansm} is an outcome for area a , team n , observed in month m , and season s . We include four main fixed effects. Because we are analyzing many years of data, the $area \times season$ fixed effect, τ_{as} , controls for the general evolution of the outcomes in area a over the years. The $month \times area$ fixed effect, κ_{am} , controls for seasonal variation in the outcomes that is specific to area a , and the $month \times record$ fixed effect, ψ_{Rm} , absorbs variation in the outcomes across the season attributable to the NFL team record, regardless of the quarterback's race. Finally, a $month \times season$ fixed effect, ϕ_{ms} , controls for changes in outcomes that are common to all areas at a particular point in time. With this set of fixed effects, the $\hat{\mu}$'s estimate the difference in how our outcomes change over the NFL season in locations with Black quarterbacks as compared to those with white quarterbacks for teams performing at the same level.

One may be concerned that the race of a team's quarterback is not randomly assigned. It is plausible that teams in areas with more racial animus are less likely to select a Black quarterback. With these specifications, if teams make this choice in response to the level of racial animus in their fan base, the $area \times season$ fixed effect controls for this selection as we are estimating within season changes. Of greater concern is if teams select Black quarterbacks when racial hate in their fan base is trending downward. This could affect our within season estimates, generating downward bias over the course of a season. In Table A5 we check for evidence of this type of selection pattern. Having a Black quarterback in season s is regressed on $Std(HateCrimeRate_{n,s-1} - HateCrimeRate_{n,s-2})$, the standardized change in the hate crime rate in the team's catchment areas over the two prior seasons. Several specifications are tested. We find no evidence that reductions in hate crimes over the prior two seasons is predictive of a team having a Black quarterback.

6.1.2 Long-run effects of local Black quarterbacks

Anti-Black hate crimes: Table 5, column 1, reports the after season estimates from equation 7 on monthly DMA anti-Black hate crimes per 100,000. We find that areas with Black quarterback teams that make it to the NFL championships experience a differentially greater reduction in the hate crime rate over the course of the NFL season as compared to areas with championship level white quarterback teams. For non-playoff Black quarterback teams, the small and statistically insignificant estimates reported in the first row implies that there is no significant or persistent difference in the hate crime rates experienced by areas with Black and white quarterbacks that do not make it to the NFL playoffs. In contrast, areas with championship level Black quarterback

teams experience a substantial reduction in the hate crime rate relative to championship white quarterback team areas. Figure 4(a) shows the evolution of the differential gap between Black and white quarterback teams for championship teams (in red) and non-playoff teams (in blue). The black line shows the average differential across all performance records. Over the course of the summer, relative hate crimes levels across these comparison groups are stable. Consistent with a response to team quarterbacks, it is only with the September start of the NFL season that the hate crime rate in championship level team areas begins to drop, persisting though the NFL season and into the spring after the season's end. In the months after the season, their hate crime rate is differentially lower by (-0.001; p-value=0.87), a 2.2% decrease.

Alternative specifications of our hate crime measure (appendix Table A6) also estimate small reductions in hate crimes for non-playoff teams and large statistically significant reductions for championship record teams. This pattern is also robust to alternative estimation approaches using different combinations of fixed effects (appendix Table A7). We find no evidence of impacts on hate crimes targeting Whites, Hispanics, or other racial groups (appendix Table A8).

Google searches for racial slurs: Table 5, column 2, row 1, and the blue line in Figure 4(b) show that when a team led by a Black quarterback fails to make the playoffs, racial slur searches increase and remain high beyond the end of the NFL season. Relative to similarly performing teams led by white quarterbacks, racial slur searches in the subsequent after season are 0.058 standard deviations (p-value = 0.050) higher in areas where a Black quarterback team did not reach the playoffs. In contrast, as shown by the red line on Figure 4(b) and the last row of Table 5, column 2, racial slur searches display a persistent decrease for championship record teams led by Black quarterbacks. Relative to areas with similarly performing teams led by white quarterbacks, racial slur searches in the subsequent season are -0.029 standard deviations (p-value = 0.61) lower in areas with championship level Black quarterback teams.

Hate speech on reddit: Table 5, column 3, reports estimates on the share of submissions removed. These estimates lack statistical precision but generally follow a similar pattern consistent with a persistent reduction in racial animus associated with championship level Black quarterback teams. Figure 4(c) plots the two-month estimates of this differential for non-playoff (in blue) and championship level teams (in red), with the average for all teams plotted in black. We see no difference in the share of submissions removed for non-playoff Black quarterback teams as

compared to non-playoff white quarterback teams. In contrast, at the championship level, the share of submissions removed is lower by -0.023 (p-value=0.13), a 22% reduction, for Black quarterback teams.

Implicit racial biases: Table 5, column 4, reports estimates on the Implicit Association Test scores of white test takers. We see no statistically significant differential changes in the IAT scores of areas with non-playoff Black quarterback teams. In contrast, teams led by a Black quarterback that reach the championship games experience a gradual decline in implicit racial biases starting in the playoff season (Figure 4(d)), reaching a reduction of -0.001 sd (p-value=0.94) that persists into the months after the end of the NFL season. Appendix Table A9 and Figure A2 examine whether exposure to Black quarterbacks might impact selection into taking an IAT test. We cannot reject the null of no differential effect on the number of IAT test takers for those exposed to championship level Black quarterback teams.

Taken together, our estimates across several different indicators of racial animus consistently suggest that in the long run, racial animus is affected by exposure to local Black quarterbacks. High performing championship level teams with Black quarterbacks lead to reductions in measures of racial animosity across a range of measures, implying that (i) local Black quarterbacks do induce racial attitudes do update, so $\lambda_L \neq 0$, and (ii) exposure to Black quarterbacks on high performing teams generate an animus reducing signal ($E[\tilde{\theta}_{ns}^L(L_{ns} = 1, Rec_{ns} = C) - \theta_{ins}] < 0$). In contrast, the long-run effects of exposure to Black quarterbacks on low performing teams that do not advance to the playoffs are generally null, with the exception of a 0.058 sd increases in google searches of racial slurs. Taken together, the presence of Black quarterbacks on these low performing teams has either a null effect or may generate a small animus increasing signal ($E[\tilde{\theta}_{ns}^L(L_{ns} = 1, Rec_{ns} = R) - \theta_{ins}] \geq 0$).

6.2 Opposition Black quarterbacks in the long-run

6.2.1 Empirical strategy

To estimate how the evolution of outcomes over the NFL season differs based on exposure to opposition Black quarterbacks we adapt the strategy developed in section 6.1. We use the same T time periods and the same *area* $\times$ *season*, *month* $\times$ *area*, *month* $\times$ *record*, and *month* $\times$ *season* fixed effects to control for the general evolution of the outcomes over the years, area specific

seasonal variation, seasonal variation attributable to NFL team performance and general time trends. Focusing on teams with white quarterbacks, we estimate

$$Y_{ansm} = \delta_0^E + \sum_T \delta_{1T}^E \mathbb{1}\{m = T\} \times E_{ns} + \tau_{as} + \kappa_{am} + \psi_{Rm} + \phi_{ms} + \epsilon_i, \quad (8)$$

where E_{ns} is a measure of exposure to opposition Black quarterbacks. We examine several different measures of exposure. For the first, we calculate G_{ns} , the number of regular season games team n played against an opposition Black quarterback in season s . With this approach, the δ_{1T}^G 's estimate the effect of one additional regular season game against an opposition Black quarterback on the outcome in period T . We also estimate a specification disaggregating G_{ns} into wins and losses against opposing Black quarterbacks. Finally, because the average effect of a single game played against an opposition Black quarterback may not generate persistent impacts, our second approach focuses on the effect of games that may be particularly salient. Focusing on the subset of teams that were in the playoffs, we generate an indicator K_{ns} which is set to one for teams who were knocked out of the playoffs by a Black quarterback. With this approach, the δ_{1T}^K 's estimate the effect of being knocked out of the playoffs by an opposition Black quarterback on the outcome in period T .

6.2.2 Long-run effects of opposition Black quarterbacks

One team's local Black quarterback is, by definition, an opposition Black quarterback to a different team. In section 6.1 we identified that championship level local Black quarterbacks generated a long-run reduction in racial animus in their teams' catchment areas. In aggregate, the long-run effects of Black quarterbacks could be quite different if this reduction was offset by persistent increases in racial animus in opposing teams' catchment areas. Table 6 and Figure 5 examine this question using several approaches to categorize exposure to opposition Black quarterbacks. Focusing on teams with white quarterbacks, panel (a) of Table 6 reports the long-run effect of playing an additional regular season game against a Black quarterback, the δ_{1T}^G 's from equation 8 estimated with G_{ns} . Panel (b) of Table 6 and the left column of Figure 5 report these estimates, differentiating between wins and losses against opposition Black quarterbacks. Finally, panel (c) of Table 6, and the right column of Figure 5 report the long-run effect of being knocked out of the playoffs by an opposition Black quarterback, the δ_{1T}^K 's from equation 8 estimated with K_{ns} .

Using these approaches, we find no evidence that facing opposing Black quarterbacks on the field leads to persistent increases in racial animus. Estimated coefficients for regular season games

in panel (a) and (b) are small in magnitude, while estimates in panel (c) lack statistical precision. None of the coefficients are statistically significant at the 95% level and we find inconsistent signs across our different indicators and for alternative measures of hate crimes (appendix Table A10). Overall, we find no solid evidence that playing against opposing Black quarterbacks increases racial animus. This is perhaps unsurprising: exposure time resulting from a single game – even a consequential one – is unlikely to produce long-lasting effects.

7 Black quarterback performance vs. game results

8 Heterogeneous analysis

How did the impacts of the presence of Black quarterbacks differ based on the racial context in which games aired? A priori, it is not clear whether media “contact” with Black quarterbacks would have larger effects in areas where people have more opportunities for direct contact with minority groups. These areas also tend to be more urban, and such contact may either strengthen the parasocial updating mechanism by making Black leadership less socially distant, or attenuate it if Black quarterbacks provide less novel information. Alternatively, effects may be larger in areas where people have fewer opportunities for direct contact with Black communities, which are often more rural, and where fewer entertainment alternatives may make NFL games especially salient. The impact may also differ by the baseline level of racial bias, measured by the 2015 Racial Animosity Index. In the short run, areas with higher baseline racial bias may be more prone to racialized backlash when a Black quarterback is associated with a negative game outcome. In the long run, positive performance by Black quarterbacks may generate stronger belief updating in high-bias areas because priors are more negative, or in low-bias areas because favorable signals may be more readily incorporated. In this section, we examine heterogeneity by the DMA-level Black population share, urban share and Racial Animosity Index.

We estimate modified versions of our baseline specifications that interact the treatment variables with a DMA characteristic, $DMACHar_j$. For each characteristic, $DMACHar_j$ is defined as an indicator equal to one for DMAs above the median and zero otherwise. The results are robust to

using the continuous versions of these characteristics. For the game-day analysis, we estimate

$$Y_{agnd} = \rho_0 + \rho_1^w Post_{dg} + \rho_2^b Post_{dg} \times 1\{QBc_{gn} = b\} \quad (9)$$

$$+ \rho_3^w Post_{dg} \times Win_{gn} + \rho_4^b Post_{dg} \times 1\{QBc_{gn} = b\} \times Win_{gn} \\ + \theta_1^w Post_{dg} \times DMACHar_j + \theta_2^b Post_{dg} \times 1\{QBc_{gn} = b\} \times DMACHar_j \quad (10)$$

$$+ \theta_3^w Post_{dg} \times Win_{gn} \times DMACHar_j \\ + \theta_4^b Post_{dg} \times 1\{QBc_{gn} = b\} \times Win_{gn} \times DMACHar_j + \eta_{ag} + \omega_d + \epsilon_{agnd}, \quad (11)$$

where $1\{QBc_{gn} = b\}$ indicates games in which at least one quarterback is Black. In this specification, θ_2^b identifies whether the differential post-game change after losses involving a Black quarterback is larger in high-characteristic DMAs than in low-characteristic DMAs. The corresponding differential for wins is given by $\theta_2^b + \theta_4^b$. For the season analysis, we analogously estimate

$$Y_{ansm} = \mu_0 + \sum_T \mu_T^r 1\{m = T\} \times L_{ns} + \sum_T \sum_{R \neq r} \mu_T^R 1\{m = T\} \times L_{ns} \times 1\{Rec_{ns} = R\} \\ + \sum_T \zeta_T^r 1\{m = T\} \times L_{ns} \times DMACHar_j \quad (12)$$

$$+ \sum_T \sum_{R \neq r} \zeta_T^R 1\{m = T\} \times L_{ns} \times 1\{Rec_{ns} = R\} \times DMACHar_j \\ + \tau_{as} + \kappa_{am} + \psi_{Rm} + \phi_{ms} + \epsilon_{ansm}, \quad (13)$$

where the ζ coefficients measure whether the season-level effects of local Black quarterbacks differ between high- and low-characteristic DMAs.

Table A11 reports the short-run heterogeneity estimates for emotionally engaging close-upset games. Overall, we find little evidence that the short-run racial retaliation effect differs systematically across these racial-context measures. For hate crimes, the estimated interaction for losses involving Black quarterbacks is positive in high-Black-share DMAs and in high-urban-share DMAs, at 0.114 and 0.0585, respectively, but both estimates are imprecisely estimated. The corresponding estimate for high-RAI DMAs is close to zero, at -0.0171 . For white IAT scores, the interaction for losses is negative for all three characteristics and statistically insignificant. The win interactions are likewise imprecisely estimated. Thus, the short-run increase in racial animus after emotionally engaging losses involving Black quarterbacks does not appear to be concentrated in a particular type of DMA.

Table A12 and Figure ?? report the corresponding long-run heterogeneity estimates. These estimates suggest some heterogeneity, but the pattern is not uniform across outcomes. For DMA Black share, championship-level Black quarterback teams generate a larger relative reduction in hate crimes in high-Black-share areas: the coefficient on *Main QB is Black* $\times$ *After season* $\times$ *In championship* $\times$ *DMAChar_j* is -0.0416 and statistically significant. However, the corresponding interaction for racial slur searches is negative but imprecisely estimated, while the IAT interaction is positive and statistically significant, indicating that the long-run pattern does not move consistently across all outcomes. For urban share, the clearest heterogeneity appears in Google searches: championship-level Black quarterback teams are associated with a larger decline in racial slur searches in more urban DMAs, with an interaction of -0.300 . The corresponding interactions for hate crimes and IAT scores are small and statistically insignificant. Finally, for baseline racial bias, we do not find evidence that high-RAI DMAs experience systematically stronger long-run reductions in racial animus. The Google-search estimates instead move in opposite directions by team performance: high-RAI areas see relatively lower racial slur searches after non-playoff Black quarterback seasons, but relatively higher searches after championship Black quarterback seasons. Figure A4 is consistent with this interpretation. The dynamic patterns do not indicate a single common form of heterogeneity across outcomes, and the clearest differences appear around championship-level Black quarterback seasons. Taken together, these results suggest that the main short-run retaliation and long-run updating effects documented above are not driven by any single racial-context group, though local context may shape the magnitude of the response for particular outcomes.

9 Star power and media visibility: Black quarterbacks vs. Black coaches

Like quarterbacks, head coaches are also consequential team leaders (Berry and Fowler (2019)). Black head coaches also occupy historically racialized leadership positions and have faced persistent barriers to entry and differential evaluation (Madden (2004), Fanning Madden and Ruther (2011), Pitts et al. (2024), Salaga and Juravich (2020)). Yet head coaches, and their performance, are not as publicly salient as that of quarterbacks. Quarterbacks receive substantial media attention both during games but also off the field through lucrative sponsorship deals and advertising visibility. They are visible in nearly every offensive play, and their success and failures are easily recognized by audiences. Head coaches, by contrast, are less directly visible as on-field actors and their

influence on game outcomes is less obvious than that of quarterbacks. They receive substantially less media attention and are not as recognizable to the general public.¹⁶

To compare these two forms of minority leadership that differ sharply in public visibility we repeat our analysis to examine the effect of the race of NFL head coaches. Table A13 reports our main short-run estimates for Black head coaches on hate crime incidents, hate speech, and IAT scores. For all three outcomes we cannot reject the null of no differential post-game change after losses involving a Black head coach, relative to losses where both head coaches are white. Across outcomes, estimates are small and statistically insignificant. The corresponding win effects, are likewise small and statistically insignificant. Table A14 reports our main long-run estimates for Black head coaches on hate crime incidents, hate speech, racial slur searches, and IAT scores. Across these outcomes, we cannot reject the null of no effect for all season performance levels.

Taken together, these null estimates suggest that leadership alone is not sufficient to generate the short-run racial retaliation and long-run updating effects that we document for Black quarterbacks. Black head coaches affect wins, losses, and season outcomes, but without comparable media salience or individualized attribution. This distinction underscores the role of the media in determining when minority leadership will influence public attitudes. Minority leaders shape racial biases and hate when they are not only in positions of authority, but are also highly visible, repeatedly discussed, and framed as individually responsible for emotionally meaningful outcomes.

10 Conclusion

This paper examines how racial animus responds to the presence and performance of minority star athletes in team sports, focusing on Black quarterbacks in the National Football League (NFL). By analyzing reactions to Black quarterbacks in NFL games, we provide new evidence on how highly visible athletes in one of America’s most prominent cultural institutions shape public attitudes and behaviors toward minority groups both around game days and over the course of a season.

First, using an event study design, we investigate how racial animus changes in the very short-run – the five days surrounding games. We find evidence of a racial retaliation effect whereby racial animus is directed toward local Black populations when a salient Black quarterback is involved – and thereby “blamable” – for the negative affect shock generated by emotionally engaging losses. Relative to games where both quarterbacks were white, losses involving a Black quarterback are

¹⁶On YouGov’s all-time sports personality fame ranking, Vince Lombardi — the legendary NFL coach and namesake of the Super Bowl trophy — is outranked by 9 NFL quarterbacks (YouGov 2026).

followed by a 18% increase in the hate crime rate. This effect is driven by emotionally engaging games – close scoring games where the loss was an upset outcome. These games experience a 43% increase in the hate crime rate. In contrast, unengaging losses – that are neither close scoring nor upset games – are associated with a much smaller, statistically insignificant increases of 9.6%. These short-run increases in racial animus after engaging losses extend beyond hate crimes. We observe a 29% increase in moderation on reddit and a 0.043 sd increase in IAT scores following such games.

Crucially, short-run emotion-driven retaliation operates alongside a long-run performance-contingent updating process in which exposure to Black quarterbacks shapes racial attitudes based on team performance. In areas with low performing Black quarterback teams that do not reach the playoffs, hate crime rates, moderation on reddit, and IAT scores are unaffected while google searches for racial slurs increase by 0.058 standard deviations. In contrast, we observe significant drops in racial animus during and after seasons where Black quarterback led teams make it to the championship games. In the months following the NFL season, these areas experience hate crime rates that are 2.2% lower, declines in racial slur searches of -0.029 standard deviations, a 22% reduction in the share of reddit submissions being moderated, and a -0.001 standard deviation decline in implicit racial biases. Finally, we find no evidence that increased exposure to opposition Black quarterbacks generates any long-run persistent changes in racial animus.

Taken together, these findings highlight two distinct mechanisms through which highly visible sporting events shape racial hostility. In the short run, emotionally engaging losses involving Black quarterbacks generate spikes in racial animus consistent with racial retaliation. Over longer horizons, however, repeated exposure to successful Black quarterbacks appears to shift racial attitudes in ways consistent with performance-based updating. These results have several implications for future work on media representation. First, researchers should consider the role of time horizons and context—including emotional responses and perceived performance—when interpreting results and assessing generalizability. Second, the transition from short-run emotional reactions to longer-run attitude updating remains understudied, suggesting a promising direction for future research.

References

Adena, M., Enikolopov, R., Petrova, M., Santarosa, V., and Zhuravskaya, E. (2015). Radio and the rise of the nazis in prewar germany. *The Quarterly Journal of Economics*, 130(4):1885–1939.

- Alesina, A. and La Ferrara, E. (2014). A test of racial bias in capital sentencing. American Economic Review, 104(11):3397–3433.
- Anderson, D. M., Crost, B., and Rees, D. I. (2020). Do economic downturns fuel racial animus? Journal of Economic Behavior & Organization, 175:9–18.
- Ang, D. (2023). The birth of a nation: Media and racial hate. American Economic Review, 113(6):1424–1460.
- Bauer, M., Cahlíková, J., Chytilová, J., Roland, G., and Želinský, T. (2023). Shifting punishment onto minorities: Experimental evidence of scapegoating. The Economic Journal, 133(652):1626–1640.
- Bayer, P., Casey, M., Ferreira, F., and McMillan, R. (2017). Racial and ethnic price differentials in the housing market. Journal of Urban Economics, 102:91–105.
- Beaman, L., Chattopadhyay, R., Duflo, E., Pande, R., and Topalova, P. (2009). Powerful women: does exposure reduce bias? The Quarterly journal of economics, 124(4):1497–1540.
- Berri, D., Farnell, A., and Simmons, R. (2023). The determinants of black quarterback pay in the national football league. Managerial and Decision Economics, 44(3):1491–1503.
- Berri, D. J. and Simmons, R. (2009). Race and the evaluation of signal callers in the national football league. Journal of Sports Economics, 10(1):23–43.
- Berry, C. R. and Fowler, A. (2019). How much do coaches matter? In Sports Analytic Conference, MIT Sloan, Boston, MA.
- Bertrand, M. and Duflo, E. (2017). Field experiments on discrimination. Handbook of economic field experiments, 1:309–393.
- Blalock, H. M. et al. (1967). Toward a theory of minority-group relations, volume 325. Wiley New York.
- Blouin, A. and Mukand, S. W. (2019). Erasing ethnicity? propaganda, nation building, and identity in rwanda. Journal of Political Economy, 127(3):1008–1062.
- Blumer, H. (1958). Race prejudice as a sense of group position. Pacific sociological review, 1(1):3–7.
- Burnett, N. J. and Van Scyoc, L. J. (2015). Compensation discrimination for defensive players: Applying quantile regression to the national football league market for linebackers and offensive linemen. Journal of Sports Economics, 16(4):375–389.
- Bursztyn, L., Egorov, G., Haaland, I., Rao, A., and Roth, C. (2022). Scapegoating during crises. In AEA Papers and Proceedings, volume 112, pages 151–155. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203.
- Cao, A., Lindo, J. M., and Zhong, J. (2023). Can social media rhetoric incite hate incidents? evidence

- from trump's "chinese virus" tweets. Journal of Urban Economics, 137:103590.
- Card, D. and Dahl, G. B. (2011). Family violence and football: The effect of unexpected emotional cues on violent behavior. The Quarterly Journal of Economics, 126(1):103–143.
- Carlana, M. (2019). Implicit stereotypes: Evidence from teachers' gender bias. The Quarterly Journal of Economics, 134(3):1163–1224.
- Caselli, M., Falco, P., and Mattera, G. (2023). When the stadium goes silent: How crowds affect the performance of discriminated groups. Journal of Labor Economics, 41(2):431–451.
- Caselli, M., Falco, P., and Somekh, B. (2024). Inside the nba bubble: how black players performed better without fans. Journal of Population Economics, 37(2):39.
- Caselli, T., Basile, V., Mitrović, J., and Granitzer, M. (2020). Hatebert: Retraining bert for abusive language detection in english. arXiv preprint arXiv:2010.12472.
- Chandrasekharan, E., Pavalanathan, U., Srinivasan, A., Glynn, A., Eisenstein, J., and Gilbert, E. (2017). You can't stay here: The efficacy of reddit's 2015 ban examined through hate speech. In Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), pages 60–73.
- Chetty, R., Hendren, N., Jones, M. R., and Porter, S. R. (2020). Race and economic opportunity in the united states: An intergenerational perspective. The Quarterly Journal of Economics, 135(2):711–783.
- Connor, P. and Evers, E. R. (2020). The bias of individuals (in crowds): Why implicit bias is probably a noisily measured individual-level construct. Perspectives on Psychological Science, 15(6):1329–1345.
- Corno, L., La Ferrara, E., and Burns, J. (2022). Interaction, stereotypes, and performance: Evidence from south africa. American Economic Review, 112(12):3848–75.
- Dee, T. S. (2005). A teacher like me: Does race, ethnicity, or gender matter? American Economic Review, 95(2):158–165.
- DellaVigna, S. and La Ferrara, E. (2015). Economic and social impacts of the media. In Handbook of media economics, volume 1, pages 723–768. Elsevier.
- Depetris-Chauvin, E., Durante, R., and Campante, F. (2020). Building nations through shared experiences: Evidence from african football. American Economic Review, 110(5):1572–1602.
- Derenoncourt, E. (2022). Can you move to opportunity? evidence from the great migration. American Economic Review, 112(2):369–408.
- Dharmapala, D. and Huq, A. (2024). Imputing unreported hate crimes using google search data.

- Journal of Law & Empirical Analysis, 1(2):232–254.
- Ducking, J., Groothuis, P. A., and Hill, J. R. (2014). Compensation discrimination in the nfl: an analysis of career earnings. Applied Economics Letters, 21(10):679–682.
- Ducking, J., Groothuis, P. A., and Hill, J. R. (2017). Compensation discrimination: An analysis of linebackers, defensive linemen, and defensive backs in the national football league. The Review of Black Political Economy, 44(3-4):251–260.
- Duncan, B., Mansour, H., and Rees, D. I. (2017). It’s just a game: The super bowl and low birth weight. Journal of Human Resources, 52(4):946–978.
- Duquennois, C. and Zeng, J. (2025). Casting roles, casting votes: Lessons from sesame street on media representation, racial biases, and voting. Working paper.
- Edmans, A., Garcia, D., and Norli, Ø. (2007). Sports sentiment and stock returns. The Journal of finance, 62(4):1967–1998.
- Egan, M., Matvos, G., and Seru, A. (2022). When harry fired sally: The double standard in punishing misconduct. Journal of Political Economy, 130(5):1184–1248.
- Eren, O. and Mocan, N. (2018). Emotional judges and unlucky juveniles. American Economic Journal: Applied Economics, 10(3):171–205.
- Fairlie, R. W., Hoffmann, F., and Oreopoulos, P. (2014). A community college instructor like me: Race and ethnicity interactions in the classroom. American Economic Review, 104(8):2567–91.
- Fanning Madden, J. and Ruther, M. (2011). Has the nfl’s rooney rule efforts “leveled the field” for african american head coach candidates? Journal of sports economics, 12(2):127–142.
- Gallo, E., Grund, T., and James Reade, J. (2013). Punishing the foreigner: implicit discrimination in the premier league based on oppositional identity. Oxford Bulletin of Economics and Statistics, 75(1):136–156.
- Gawronski, B. (2019). Six lessons for a cogent science of implicit bias and its criticism. Perspectives on Psychological Science, 14(4):574–595.
- Gershenson, S., Hart, C. M., Hyman, J., Lindsay, C. A., and Papageorge, N. W. (2022). The long-run impacts of same-race teachers. American Economic Journal: Economic Policy, 14(4):300–342.
- Glover, D., Pallais, A., and Pariente, W. (2017). Discrimination as a self-fulfilling prophecy: Evidence from french grocery stores. The Quarterly Journal of Economics, 132(3):1219–1260.
- Greenwald, A. G., Poehlman, T. A., Uhlmann, E. L., and Banaji, M. R. (2009). Understanding and using the Implicit Association Test: III. Meta-analysis of predictive validity. Journal of Personality and Social Psychology, 97(1):17.

- Gregory-Smith, I., Bryson, A., and Gomez, R. (2023). Discrimination in a rank order contest: Evidence from the nfl draft. Journal of Economics, Race, and Policy, 6(3):215–231.
- Hartvigsen, T., Gabriel, S., Palangi, H., Sap, M., Ray, D., and Kamar, E. (2022). Toxigen: A large-scale machine-generated dataset for adversarial and implicit hate speech detection. arXiv preprint arXiv:2203.09509.
- Helman, E., Calanchini, J., Flake, J. K., and Leitner, J. B. (2019). Establishing construct validity evidence for regional measures of explicit and implicit racial bias. Journal of Experimental Psychology: General, 148(6):1022.
- Ivandić, R., Kirchmaier, T., Saeidi, Y., and Blas, N. T. (2024). Football, alcohol, and domestic abuse. Journal of public economics, 230:105031.
- Karp, A. (2014). Sports business journal: Nfl teams saw local market ratings gain in ‘13: Chiefs,seahawks up big. Accessed: 2025-11-28.
- Keefer, Q. A. and Kniesner, T. J. (2023). Injury risk, concussions, race, and pay in the nfl. Journal of Risk and Uncertainty, 67(2):107–136.
- Keefer, Q. A. W. (2013). Compensation discrimination for defensive players: Applying quantile regression to the national football league market for linebackers. Journal of Sports Economics, 14(1):23–44.
- Kline, P., Rose, E. K., and Walters, C. R. (2022). Systemic discrimination among large us employers. The Quarterly Journal of Economics, 137(4):1963–2036.
- Lang, K. and Kahn-Lang Spitzer, A. (2020). Race discrimination: An economic perspective. Journal of Economic Perspectives, 34(2):68–89.
- Lang, K., Manove, M., and Dickens, W. T. (2005). Racial discrimination in labor markets with posted wage offers. American Economic Review, 95(4):1327–1340.
- Lim, J. and Meer, J. (2020). Persistent effects of teacher–student gender matches. Journal of Human Resources, 55(3):809–835.
- Lowes, S., Nunn, N., Robinson, J. A., and Weigel, J. (2015). Understanding ethnic identity in africa: Evidence from the implicit association test (iat). American Economic Review, 105(5):340–345.
- Madden, J. F. (2004). Differences in the success of nfl coaches by race, 1990-2002: Evidence of last hire, first fire. Journal of Sports Economics, 5(1):6–19.
- Marble, W., Mousa, S., Siegel, A. A., et al. (2021). Can exposure to celebrities reduce prejudice? the effect of mohamed salah on islamophobic behaviors and attitudes. American Political Science Review, 115(4):1111–1128.

- Marquez-Velarde, G., Grashow, R., Glass, C., Blaschke, A. M., Gillette, G., Taylor, H. A., and Whittington, A. J. (2023). The paradox of integration: racial composition of nfl positions from 1960 to 2020. Sociology of Race and Ethnicity, 9(4):451–469.
- Matsuzawa, K. and Arnesen, E. (2024). Sports betting legalization amplifies emotional cues & intimate partner violence. Available at SSRN.
- Miller, P. R., Flores, A. R., Haider-Markel, D. P., Lewis, D. C., Tadlock, B., and Taylor, J. K. (2020). The politics of being “cait”: Caitlyn jenner, transphobia, and parasocial contact effects on transgender-related political attitudes. American politics research, 48(5):622–634.
- Montolio, D. and Planells-Struse, S. (2016). How time shapes crime: The temporal impacts of football matches on crime. Regional Science and Urban Economics, 61:99–113.
- Mulholland, S. E. (2013). White supremacist groups and hate crime. Public Choice, 157(1):91–113.
- Müller, K. and Schwarz, C. (2021). Fanning the flames of hate: Social media and hate crime. Journal of the European Economic Association, 19(4):2131–2167.
- Müller, K. and Schwarz, C. (2023). From hashtag to hate crime: Twitter and antiminority sentiment. American Economic Journal: Applied Economics, 15(3):270–312.
- Munyo, I. and Rossi, M. A. (2013). Frustration, euphoria, and violent crime. Journal of Economic Behavior & Organization, 89:136–142.
- Ourand, J. (2022). Sports business journal: Gaining yardage: Local tv ratings rise for 21 of nfl’s 32 teams including key los angeles market. Accessed: 2025-11-28.
- Paluck, E. L. (2009). Reducing intergroup prejudice and conflict using the media: a field experiment in rwanda. Journal of personality and social psychology, 96(3):574.
- Paluck, E. L., Porat, R., Clark, C. S., and Green, D. P. (2021). Prejudice reduction: Progress and challenges. Annual review of psychology, 72:533–560.
- Parsons, C. A., Sulaeman, J., Yates, M. C., and Hamermesh, D. S. (2011). Strike three: Discrimination, incentives, and evaluation. American economic review, 101(4):1410–1435.
- Pitts, J. D., Evans, B., and Johnson, J. D. (2024). Race and the probability of becoming a head coach for nfl coordinators since the introduction of the rooney rule: why isn’t eric bieniemy a head coach yet? The Review of Black Political Economy, 51(1):17–43.
- Plant, E. A., Devine, P. G., Cox, W. T., Columb, C., Miller, S. L., Goplen, J., and Peruche, B. M. (2009). The obama effect: Decreasing implicit prejudice and stereotyping. Journal of Experimental Social Psychology, 45(4):961–964.
- Price, J. and Wolfers, J. (2010). Racial discrimination among nba referees. The Quarterly journal of

- economics, 125(4):1859–1887.
- Ratliff, K. A. and Smith, C. T. (2021). Lessons from two decades of Project Implicit. In The Cambridge handbook of implicit bias and racism, page 4. Cambridge University Press.
- Reddit (2020). Reddit content policy. <https://www.redditinc.com/policies/content-policy>. Accessed 2024.
- Rees, D. I. and Schnepel, K. T. (2009). College football games and crime. Journal of Sports Economics, 10(1):68–87.
- Reid, J. (2022). The Rise of the Black Quarterback: What It Means for America. Disney Electronic Content.
- Rhoden, W. C. (2007). Third and a Mile: The Trials and Triumph of the Black Quarterback. ESPN Books.
- Riley, E. (2024). Role models in movies: The impact of queen of katwe on students' educational attainment. Review of Economics and Statistics, 106(2):334–351.
- Rosenzweig, L. R. and Zhou, Y.-Y. (2021). Team and nation: Sports, nationalism, and attitudes toward refugees. Comparative Political Studies, 54(12):2123–2154.
- Rutland, A. and Killen, M. (2015). A developmental science approach to reducing prejudice and social exclusion: Intergroup processes, social-cognitive development, and moral reasoning. Social Issues and Policy Review, 9(1):121–154.
- Sakong, J. (2026). Do black electoral victories shift racial bias? evidence from close elections. Journal of Public Economics, 256:105594.
- Salaga, S. and Juravich, M. (2020). National football league head coach race, performance, retention, and dismissal. Sport Management Review, 23(5):978–991.
- Sarsons, H. (2017). Interpreting signals in the labor market: evidence from medical referrals. Job market paper, pages 141–145.
- Schiappa, E., Gregg, P. B., and Hewes, D. E. (2005). The parasocial contact hypothesis. Communication Monographs, 72(1):92–115.
- Schimmack, U. (2021). The Implicit Association Test: A method in search of a construct. Perspectives on Psychological Science, 16(2):396–414.
- Skiba, R. J., Michael, R. S., Nardo, A. C., and Peterson, R. L. (2002). The color of discipline: Sources of racial and gender disproportionality in school punishment. The urban review, 34(4):317–342.
- Stephens-Davidowitz, S. (2014). The cost of racial animus on a black candidate: Evidence using google search data. Journal of Public Economics, 118:26–40.

- Stoecker, C., Sanders, N. J., and Barreca, A. (2016). Success is something to sneeze at: Influenza mortality in cities that participate in the super bowl. American Journal of Health Economics, 2(1):125–143.
- Volz, B. D. (2017). Race and quarterback survival in the national football league. Journal of Sports Economics, 18(8):850–866.
- Yanagizawa-Drott, D. (2014). Propaganda and Conflict: Evidence from the Rwandan Genocide *. The Quarterly Journal of Economics, 129(4):1947–1994.
- YouGov (2026). The most famous all-time sports personalities in america. YouGov Ratings. Data collection period: First quarter of 2026. Accessed April 20, 2026.

11 Tables

Table 1: Summary statistics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel a: Game day data									
	Units	Obs. level	Weights	Year of game	Obs.	Games x teams	Season x teams	Mean	Std. Dev
Anti-Black hate crimes	Crimes per 10 mil.	Date x DMA	DMA population	2002 - 2025	306625	9709	729	0.165	2.011
IAT score	Std. dev.	Individual tests		2003 - 2024	513614	6019	664	0.176	0.952
Share removed	Share	Date x Subreddit	Submission count	2015 - 2024	19734	4002	316	0.120	0.115
Appendix outcomes									
Submissions	Count	Date x Subreddit		2015 - 2024	19805	4002	316	65	83
Anti-Black hate speech	ToxiGen score	Date x Subreddit	Submission count	2015 - 2024	19664	4002	316	0.00008	0.00164
IAT count	Count	Date x DMA		2002 - 2025	4575065	9709	729	0.2	1.2
Panel b: Season data									
	Units	Obs. level	Weights	Complete seasons	Obs.		Season x teams	Mean	Std. Dev
Hate crimes	Crimes per 100,000	Month x DMA	DMA population	2002 - 2023	52044		682	0.053	0.134
Google slur searches	Std. index	2 Months x DMA	DMA population	2004 - 2023	15432		613	-0.150	0.811
IAT score	Std. index	Individuals tests		2004 - 2023	3584169		618	0.168	0.949
Share removed	Share	Month x Subreddit	Submission count	2016 - 2023	3024		252	0.103	0.072
Appendix outcomes									
Submissions	Count	Month x Subreddit		2016 - 2023	3024		252	1097	969
Anti-Black hate speech	ToxiGen score	Month x Subreddit	Submission count	2016 - 2023	3012		252	0.00007	0.00038
IAT count	Count	Month x DMA		2002 - 2024	863604		736	4	22

Table 2: Game day impacts on hate crimes

	(1) All	(2) Close-upsets	(3) Either	(4) Neither
White-white games:				
<i>Post</i>	-0.00450 (0.0168)	-0.0121 (0.0368)	-0.00225 (0.0209)	-0.00645 (0.0275)
<i>Post × Win</i>	-0.00122 (0.0115)	0.0348 (0.0297)	-0.00311 (0.0154)	0.000879 (0.0174)
Any Black quarterback games:				
<i>Post × Any Black QB</i>	0.0366*** (0.0130)	0.0863*** (0.0324)	0.0485*** (0.0171)	0.0191 (0.0200)
<i>Post × Any Black QB × Win</i>	-0.0214 (0.0179)	-0.104** (0.0469)	-0.0294 (0.0238)	-0.00944 (0.0273)
$\rho_2^b + \rho_4^b$	0.015	-0.018	0.019	0.010
Dependent mean	0.200	0.202	0.201	0.199
Observations	306625	44715	172405	134220
Game × teams	9709	1401	5471	4238

Note: Standard errors are reported in parentheses, with the following significance indicators:
 * p<0.1, ** p<0.05 and *** p<0.01.

Table 3: Game day impacts on hate speech indicators and IAT scores

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Submission removals				White IAT scores			
	All	Close-upsets	Either	Neither	All	Close-upsets	Either	Neither
White-white games:								
<i>Post</i>	0.0596*** (0.00852)	0.0346 (0.0232)	0.0598*** (0.0116)	0.0553*** (0.0107)	-0.0124 (0.0127)	0.00837 (0.0287)	-0.00589 (0.0171)	-0.0235 (0.0175)
<i>Post × Win</i>	-0.0373*** (0.00539)	-0.0234 (0.0149)	-0.0401*** (0.00759)	-0.0328*** (0.00719)	-0.00740 (0.00837)	0.0169 (0.0218)	-0.00499 (0.0114)	-0.0101 (0.0122)
Any Black quarterback games:								
<i>Post × Any Black QB</i>	0.000160 (0.00651)	0.0455** (0.0177)	0.00453 (0.00918)	-0.00545 (0.00865)	0.0000162 (0.00930)	0.0430* (0.0228)	0.00719 (0.0124)	-0.00907 (0.0140)
<i>Post × Any Black QB × Win</i>	-0.00511 (0.00803)	-0.0395* (0.0215)	-0.00567 (0.0112)	-0.00514 (0.0111)	-0.00166 (0.0133)	-0.0743** (0.0338)	-0.0141 (0.0178)	0.0152 (0.0199)
$\rho_2^b + \rho_4^b$	-0.005	0.006	-0.001	-0.011	-0.002	-0.031	-0.007	0.006
Dependent mean	0.145	0.155	0.147	0.143	0.176	0.188	0.177	0.174
Observations	19701	3009	11252	8449	513614	73129	286180	227434
Game × teams	3960	605	2260	1700	6019	883	3363	2656

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table 4: Engaging game day impacts for local and opposition Black quarterbacks

	(1)	(2)	(3)
	Hate crime rate	Sub. removals	IAT scores
Local Black quarterback games:			
<i>Post × Only local QB is Black</i>	0.0951** (0.0401)	0.0530** (0.0217)	0.0340 (0.0242)
<i>Post × Only local QB is Black × Win</i>	-0.0816 (0.0567)	-0.0543** (0.0271)	-0.0752* (0.0404)
Opposition Black quarterback games:			
<i>Post × Only opp. QB is Black</i>	0.0757* (0.0395)	0.0366* (0.0196)	0.0540 (0.0335)
<i>Post × Only opp. QB is Black × Win</i>	-0.124** (0.0613)	-0.0261 (0.0252)	-0.0761* (0.0456)
P-value of $\rho_2^l = \rho_2^o$	.68	.46	.57
P-value of $\rho_2^l + \rho_4^l = \rho_2^o + \rho_4^o$	.26	.58	.63
Dependent mean	0.179	0.126	0.188
Observations	44715	3009	73129
Game × teams	1401	605	883

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table 5: Season effects of local Black quarterbacks by team performance

	(1) Hate crime rate (monthly DMA)	(2) Google slur index (Standardized)	(3) Submission removals (Share)	(4) IAT scores (Standardized)
Not in playoffs				
<i>Main QB is Black × After season</i>	-0.000198 (0.00351)	0.0576** (0.0293)	0.00613 (0.0141)	0.00213 (0.00563)
In playoffs only				
<i>Main QB is Black × After season × In playoffs only</i>	-0.00561 (0.00700)	-0.0533 (0.0464)	-0.00205 (0.0152)	-0.00109 (0.0101)
In championships				
<i>Main QB is Black × After season × In championship</i>	-0.0175* (0.00921)	-0.140*** (0.0538)	0.00335 (0.0424)	-0.0421*** (0.0140)
Compared to white QB championship teams:	-0.018**	-0.082*	0.009	-0.040***
Dep. mean	0.0530	-0.151	0.103	0.168
Observations	52044	15414	2927	3584160
Team × seasons	682	613	244	618

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table 6: Season effects of opposition quarterback's race

	(1) Hate crimes (monthly DMA)	(2) Google slur index	(3) Submission removals (Share)	(4) IAT scores (Standardized)
Panel a: Effects of opposition quarterback games in the regular season				
<i>Games against Opp. Black QBs × After season</i>	0.00216** (0.00105)	-0.000819 (0.00626)	-0.000663 (0.00208)	0.000597 (0.00136)
Dependent mean	0.0537	-0.132	0.0970	0.180
Observations	40080	11424	2147	2671482
Panel b: Effects of opposition quarterback games in the regular season by game outcome				
<i>Losses against Opp. Black QBs × After season</i>	0.00103 (0.00175)	-0.00957 (0.00853)	-0.000637 (0.00287)	-0.00108 (0.00196)
<i>Wins against Opp. Black QBs × After season</i>	0.00315** (0.00122)	0.00843 (0.00857)	-0.000524 (0.00247)	0.00254 (0.00180)
Dependent mean	0.0537	-0.132	0.0970	0.180
Observations	40080	11424	2147	2671482
Panel c: Effects of being knocked out by an opposition Black quarterback team				
<i>Knockout QB is Black × After season</i>	-0.0148* (0.00830)	0.00960 (0.0522)	-0.0113 (0.0214)	-0.00503 (0.00956)
Dependent mean	0.0511	-0.0845	0.106	0.176
Observations	13776	3888	671	902888

Note: Standard errors are reported in parentheses, with the following significance indicators: * $p < 0.1$, ** $p < 0.05$ and *** $p < 0.01$.

12 Figures

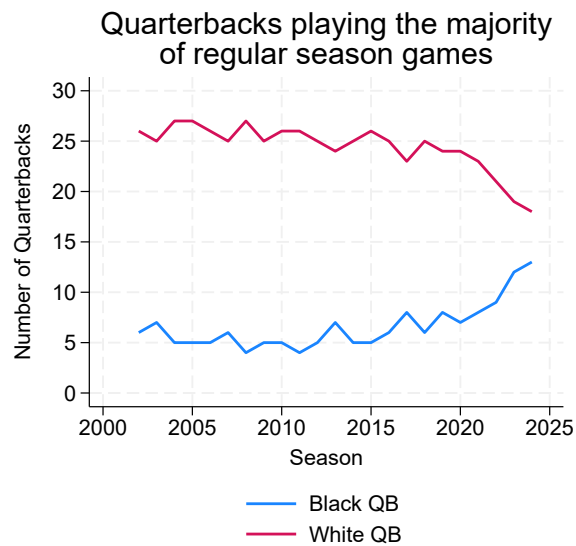


Figure 1: Black quarterbacks in the NFL

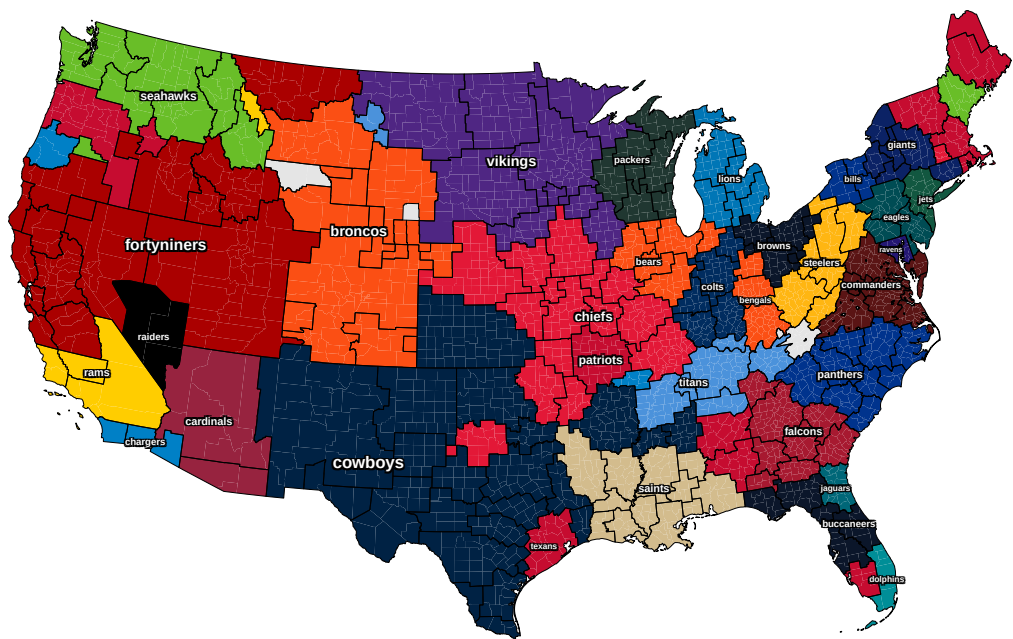


Figure 2: NFL team catchment areas

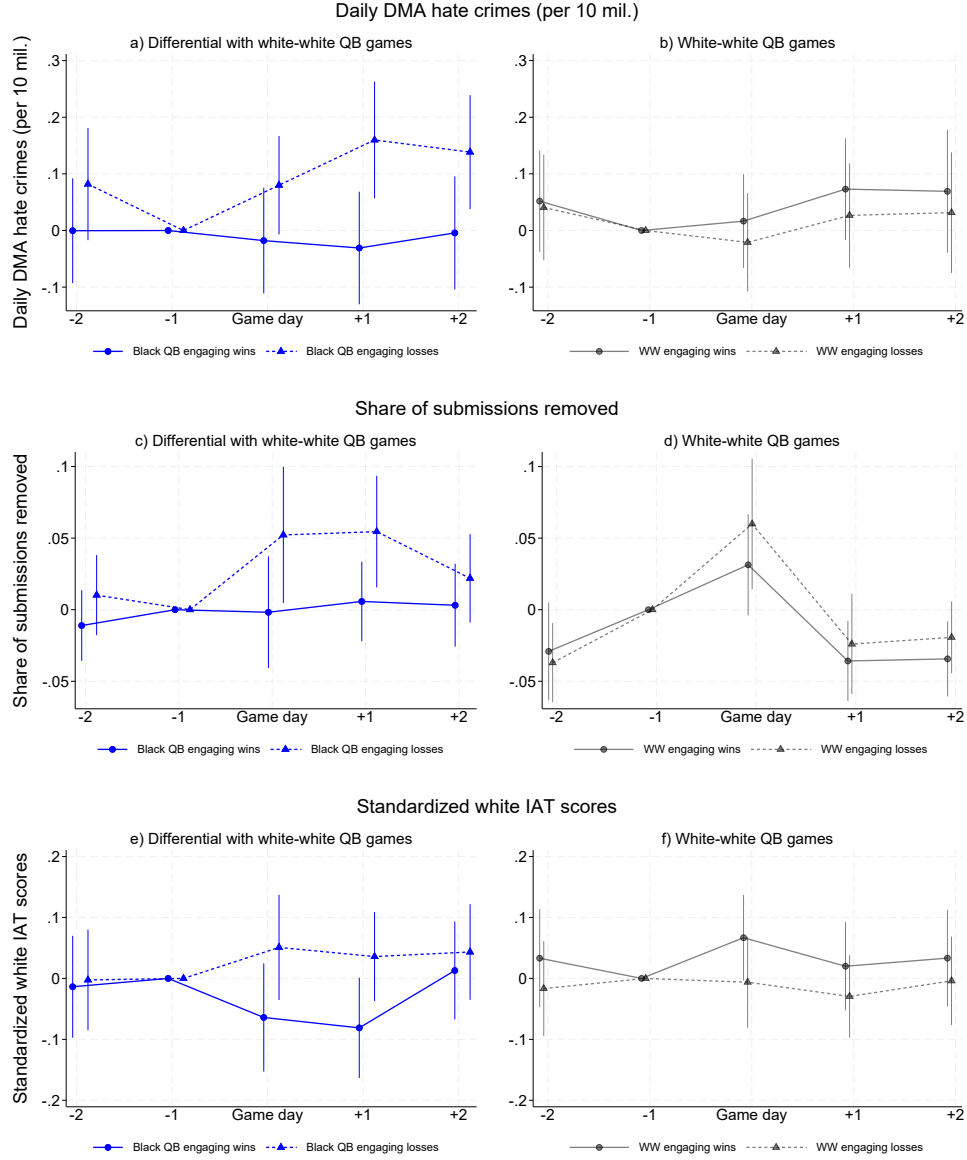


Figure 3: Game day effects of Black QBs

Notes: Figures on the left plot the differential effect of Black quarterbacks for engaging games as estimated by equation 6. The ρ_2^b estimates are plotted for engaging losses and the $(\rho_2^b + \rho_4^b)$ estimates for engaging wins. Figures on the right plot the ρ_1^w estimates and the $(\rho_1^w + \rho_3^w)$ estimates showing the effects of engaging losses and wins around white-white quarterback games.

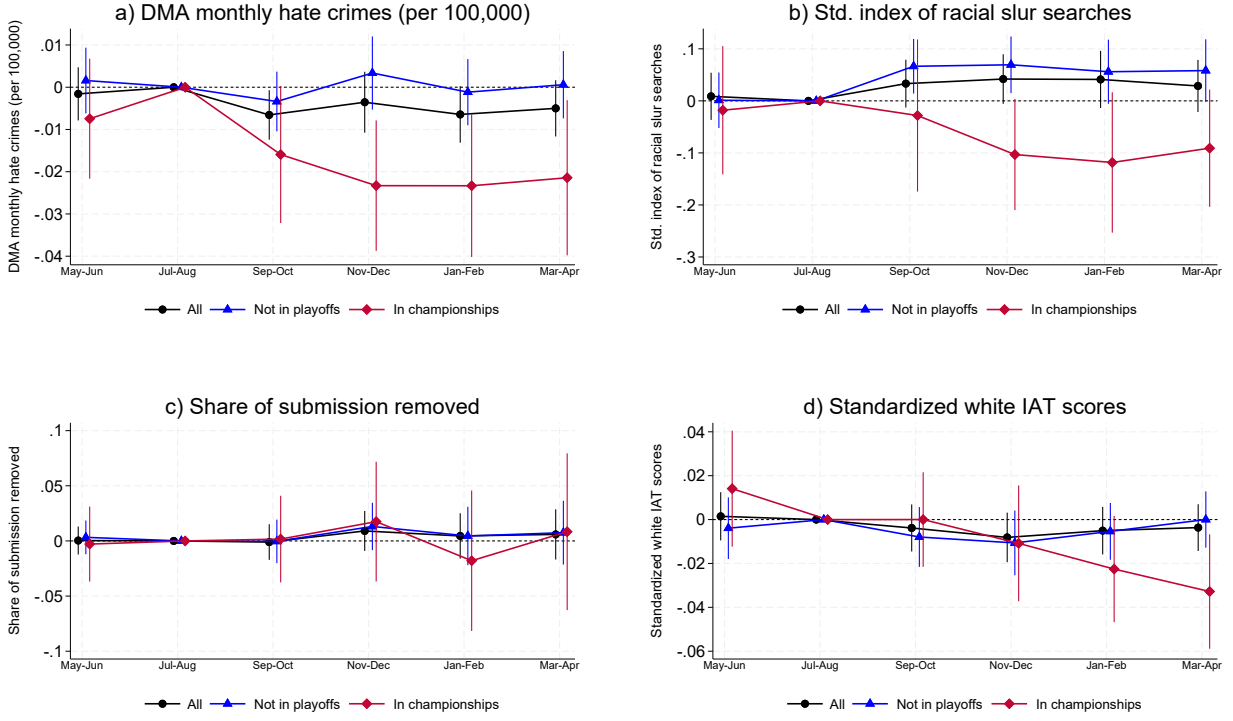


Figure 4: Long-run effects of local Black QBs

Notes: Figures plot the μ_T^r and μ_T^c from equation 7 in blue and red respectively. Aggregate estimates for all performance levels are plotted in black. These estimates control for $location \times season$, $month \times location$, $month \times record$, and $month \times season$ fixed effects.

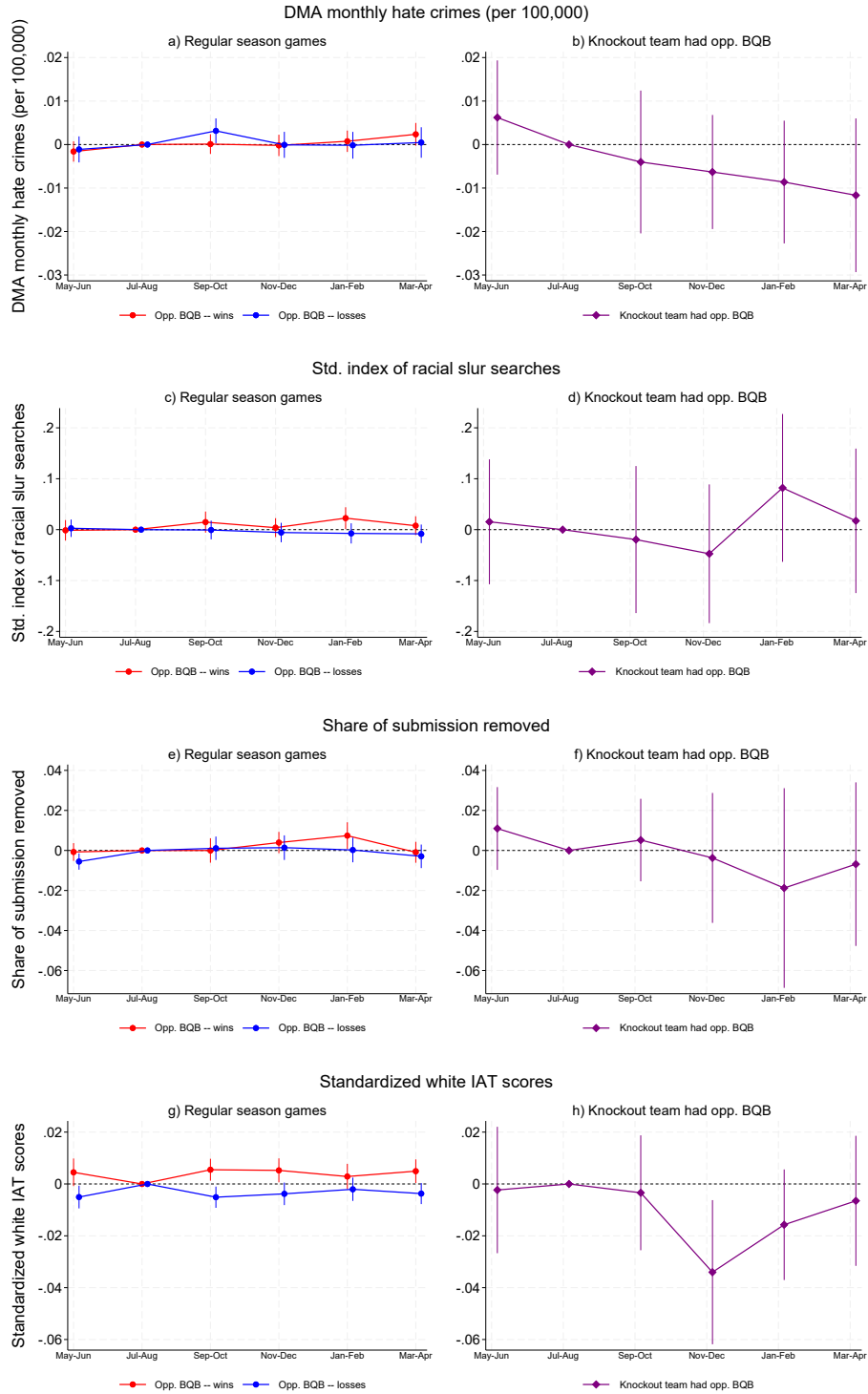


Figure 5: Long-run effects of opposition Black QBs

Notes: For all figures, the sample is limited to team-seasons where the main quarterback was white. Figures on the left plot the differential effect of an additional regular season game played against opposition Black quarterbacks. Figures on the right plot the differential effect of being knocked out of the playoffs by a team with an Black quarterback.

Online Appendix

A1 Appendix Tables

Table A1: Robustness to alternative outcome measures of daily DMA hate crimes

	Main (incidents per 10 mil.)	Extensive indicator (1 if has incidents)	Zero adjusted log log(incidents+1)	Linear raw (incident count)	Poisson raw (incident count)	Unweighted main
Panel a: All games						
<i>Post × Any Black QB</i>	0.0366*** (0.0130)	0.00439** (0.00171)	0.00357*** (0.00129)	0.00573*** (0.00202)	0.177*** (0.0646)	0.0716*** (0.0207)
<i>Post × Any Black QB × Win</i>	-0.0214 (0.0179)	-0.00224 (0.00231)	-0.00206 (0.00172)	-0.00355 (0.00269)	-0.105 (0.0903)	-0.0279 (0.0298)
Dependent mean	0.200	0.027	0.020	0.030	0.288	0.165
Observations	306625	306625	306625	306625	31755	306625
Panel b: Close upset games						
<i>Post × Any Black QB</i>	0.0863*** (0.0324)	0.0107*** (0.00404)	0.00798*** (0.00298)	0.0121*** (0.00455)	0.464*** (0.172)	0.171*** (0.0596)
<i>Post × Any Black QB × Win</i>	-0.104** (0.0469)	-0.0127** (0.00620)	-0.00958** (0.00455)	-0.0148** (0.00694)	-0.544** (0.233)	-0.199** (0.0773)
Dependent mean	0.202	0.027	0.020	0.030	0.281	0.179
Observations	44715	44715	44715	44715	4710	44715
Panel c: Neither games						
<i>Post × Any Black QB</i>	0.0191 (0.0200)	0.00257 (0.00266)	0.00209 (0.00203)	0.00324 (0.00323)	0.0783 (0.0973)	0.0283 (0.0335)
<i>Post × Any Black QB × Win</i>	-0.00944 (0.0273)	-0.000909 (0.00349)	-0.000963 (0.00263)	-0.00185 (0.00415)	-0.0380 (0.139)	0.00982 (0.0436)
Dependent mean	0.199	0.027	0.020	0.030	0.291	0.160
Observations	134220	134220	134220	134220	13800	134220

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A2: Robustness to fixed effects

	(1)	(2)	(3)	(4)
Panel a: All games				
<i>Post</i> × <i>Any Black QB</i>	0.036559*** (0.013010)	0.036831*** (0.013007)	0.036837*** (0.013007)	0.036836*** (0.013011)
<i>Post</i> × <i>Any Black QB</i> × <i>Win</i>	-0.021396 (0.017938)	-0.021346 (0.017943)	-0.021355 (0.017943)	-0.021353 (0.017949)
Dependent mean	0.200	0.200	0.200	0.200
Observations	306625	306625	306625	306625
Panel b: Close upset games				
<i>Post</i> × <i>Any Black QB</i>	0.086278*** (0.032374)	0.086517*** (0.032255)	0.086524*** (0.032255)	0.086521*** (0.032330)
<i>Post</i> × <i>Any Black QB</i> × <i>Win</i>	-0.104207** (0.046903)	-0.103197** (0.046863)	-0.103204** (0.046863)	-0.103203** (0.046972)
Dependent mean	0.202	0.202	0.202	0.202
Observations	44715	44715	44715	44715
Panel c: Neither games				
<i>Post</i> × <i>Any Black QB</i>	0.019148 (0.019986)	0.019660 (0.019993)	0.019665 (0.019993)	0.019663 (0.020008)
<i>Post</i> × <i>Any Black QB</i> × <i>Win</i>	-0.009443 (0.027349)	-0.009405 (0.027348)	-0.009411 (0.027348)	-0.009410 (0.027369)
Dependent mean	0.199	0.199	0.199	0.199
Observations	134220	134220	134220	134220
FE: DMA × Game-team	Yes	Yes	No	No
FE: Day of the week	Yes	No	No	No
FE: DMA	.	.	No	Yes
FE: Game-team	.	.	Yes	Yes

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A3: Game day impacts on other hate crimes

	(1) Anti-Black	(2) Anti-white	(3) Anti-Hispanic	(4) Anti-other race
White-white games:				
<i>Post</i>	-0.0121 (0.0368)	-0.0216 (0.0227)	0.0126 (0.0175)	-0.00224 (0.0230)
<i>Post</i> × <i>Win</i>	0.0348 (0.0297)	0.000238 (0.0160)	0.0300** (0.0135)	-0.00334 (0.0191)
Any Black quarterback games:				
<i>Post</i> × <i>Any Black QB</i>	0.0863*** (0.0324)	0.00797 (0.0178)	0.00525 (0.0138)	-0.000923 (0.0199)
<i>Post</i> × <i>Any Black QB</i> × <i>Win</i>	-0.104** (0.0469)	0.00741 (0.0253)	-0.0450** (0.0212)	0.0230 (0.0273)
$\rho_2^b + \rho_4^b$	-0.018	0.015	-0.040**	0.022
Dependent mean	0.202	0.061	0.044	0.066
Observations	44715	44715	44715	44715
Game × teams	1401	1401	1401	1401

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A4: Game day impacts on submission numbers, anti-Black speech and IAT test taking – Close-upset games

	(1) Submissions	(2) Anti-Black speech	(3) IAT test takers
White-white games:			
<i>Post</i>	90.59*** (11.23)	0.00000669 (0.0000360)	0.0152 (0.0149)
<i>Post</i> × <i>Win</i>	16.93** (6.748)	-0.000206 (0.000182)	-0.0222 (0.0136)
Any Black quarterback games:			
<i>Post</i> × <i>Any Black QB</i>	-4.729 (5.651)	0.0000999 (0.000126)	-0.0206 (0.0127)
<i>Post</i> × <i>Any Black QB</i> × <i>Win</i>	12.15 (11.93)	0.000127 (0.000223)	0.0251 (0.0179)
Win test	7.42	0.000	0.005
Dependent mean	68.543	0.000	0.155
Observations	3025	2997	660965
Game × teams	609	602	1401

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A5: Changes in hate crimes do not predict quarterback race

	(1)	(2)	(3)	(4)
Main QB is Black				
Std. change in hate crimes from s-2 to s-1	-0.0219 (0.0181)	-0.00544 (0.0154)	-0.00612 (0.0152)	-0.00907 (0.0140)
s-1 season QB was Black		0.457*** (0.0430)	0.365*** (0.0485)	
s-2 season QB was Black		0.137*** (0.0432)	0.120*** (0.0428)	
s-1 season made championships			-0.0621 (0.0478)	
s-1 season made playoffs only			-0.0408 (0.0354)	
s-1 season made championships with Black QB			0.437*** (0.114)	
s-1 season made playoffs only with Black QB			0.216*** (0.0786)	
Observations	586	586	586	438
FE: Team	Yes	Yes	Yes	Yes
FE: Season	Yes	Yes	Yes	Yes
Subsample: Had a white QB in s-1 and s-2				Yes

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A6: Season effects of local Black quarterbacks – robustness

	(1) Main (DMA/mo/100k)	(2) Extensive indicator (1 if has incidents)	(3) Zero adjusted log log(incidents+1)	(4) Linear raw (incident count)	(5) Poisson raw (incident count)	(6) Unweighted main
Not in playoffs						
<i>Main QB is Black × After season</i>	-0.000198 (0.00351)	-0.0137 (0.0170)	-0.0144 (0.0165)	-0.0175 (0.0569)	0.0128 (0.0553)	-0.0111** (0.00533)
In playoffs only						
<i>Main QB is Black × After season × In playoffs only</i>	-0.00561 (0.00700)	-0.0108 (0.0273)	-0.00458 (0.0307)	-0.0915 (0.108)	-0.0521 (0.0992)	-0.00184 (0.00916)
In championships						
<i>Main QB is Black × After season × In championship</i>	-0.0175* (0.00921)	-0.0317 (0.0412)	-0.0535 (0.0441)	-0.201 (0.139)	-0.273** (0.136)	-0.00901 (0.0133)
Compared to white QB championship teams:	-0.018**	-0.045	-0.068*	-0.218*	-0.260**	-0.020*
Dependent mean	0.063	0.335	0.374	0.939	1.265	0.053
Observations	52044	52044	52044	52044	38629	52044

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A7: Season estimates for local Black quarterbacks – Robustness to fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Hate crime rate						
Not in playoffs							
<i>After season</i>		-0.00922*** (0.00219)	-0.00768*** (0.00188)				
<i>Main QB is Black × After season</i>	-0.000198 (0.00351)	0.000623 (0.00420)	-0.00107 (0.00319)	-0.000811 (0.00331)	-0.00290 (0.00344)	0.00125 (0.00356)	-0.000520 (0.00423)
In playoffs only							
<i>After season × In playoffs only</i>		0.00251 (0.00367)	-0.000737 (0.00328)				
<i>Main QB is Black × After season × In playoffs only</i>	-0.00561 (0.00700)	-0.00283 (0.00734)	0.00137 (0.00568)	-0.00316 (0.00569)	0.000235 (0.00598)	-0.00714 (0.00658)	-0.000619 (0.00760)
In championships							
<i>After season × In championship</i>		0.00913** (0.00454)	0.00194 (0.00317)				
<i>Main QB is Black × After season × In championship</i>	-0.0175* (0.00921)	-0.0209** (0.00920)	-0.0104 (0.00640)	-0.00951 (0.00644)	-0.00649 (0.00732)	-0.0174** (0.00809)	-0.0223** (0.00980)
Compared to white QB championship teams:	-0.018**	-0.020**	-0.011**	-0.010*	-0.009	-0.016**	-0.023**
Observations	52044	52044	52044	52044	52044	52044	52044
FE: DMA × season	Yes	Yes	No	Yes	Yes	Yes	Yes
FE: Month × season	Yes	No	No	Yes	Yes	Yes	No
FE: DMA × month	Yes	No	No	No	Yes	No	Yes
FE: Record × month	Yes	No	No	No	No	Yes	Yes
FE: DMA	.	.	Yes	.	.	.	.
FE: Season	.	.	Yes	.	.	.	.

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A8: Season effects of local Black quarterbacks by team performance

	(1) Anti-Black	(2) Anti-White	(3) Anti-Hispanic	(4) Anti-Other race
Not in playoffs				
<i>Main QB is Black × After season</i>	-0.000198 (0.00351)	-0.000459 (0.00199)	-0.000603 (0.00150)	-0.00324 (0.00239)
In playoffs only				
<i>Main QB is Black × After season × In playoffs only</i>	-0.00561 (0.00700)	-0.00132 (0.00387)	0.00262 (0.00266)	0.000275 (0.00391)
In championships				
<i>Main QB is Black × After season × In championship</i>	-0.0175* (0.00921)	0.000919 (0.00456)	0.00212 (0.00318)	0.00178 (0.00500)
Dep. mean				
Observations	52044	52044	52044	52044

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A9: Season effects of local Black quarterbacks by team performance – appendix outcomes

	(1) Submissions	(2) Anti-Black hate speech	(3) IAT tests
Not in playoffs			
<i>Main QB is Black × After season</i>	-88.40 (204.3)	0.0000900 (0.0000707)	-0.0795 (0.229)
In playoffs only			
<i>Main QB is Black × After season × In playoffs only</i>	-97.96 (200.9)	-0.000149 (0.0000992)	-0.261 (0.479)
In championships			
<i>Main QB is Black × After season × In championship</i>	60.63 (354.4)	0.000329 (0.000311)	0.995* (0.588)
Compared to white QB championship teams:	-27.8	0.000	0.916
Dep. mean	1102.5	0	4.610
Observations	2927	2916	816024
Team × seasons	244	244	713

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A10: Season effects of opposition Black quarterbacks – alternative measures of hate crimes

	(1) Main (DMA/mo/100k)	(2) Extensive indicator (1 if has incidents)	(3) Zero adjusted log log(incidents+1)	(4) Linear raw (incident count)	(5) Poisson raw (incident count)	(6) Unweighted main
Panel a: Effects of opposition quarterback games in the regular season						
<i>Games against Opp. Black QBs × After season</i>	0.00216** (0.00105)	0.00379 (0.00559)	0.0196** (0.00968)	-0.0181 (0.149)	0.0345** (0.0138)	0.00210 (0.00148)
Dependent mean	0.054	0.338	0.380	0.970	1.322	0.054
Observations	40080	40080	40080	40080	29407	40080
Panel b: Effects of opposition quarterback games in the regular season						
<i>Losses against Opp. Black QBs × After season</i>	0.00103 (0.00175)	-0.00231 (0.00751)	0.000886 (0.0147)	-0.266 (0.286)	0.0146 (0.0206)	0.00182 (0.00218)
<i>Wins against Opp. Black QBs × After season</i>	0.00315** (0.00122)	0.0102 (0.00684)	0.0378*** (0.0114)	0.218* (0.125)	0.0529*** (0.0177)	0.00233 (0.00188)
Dependent mean	0.0537	0.338	0.380	0.970	1.322	0.0537
Observations	40080	40080	40080	40080	29407	40080
Panel c: Effects of being knocked out by an opposition Black quarterback team						
<i>Knockout QB is Black × After season</i>	-0.0148* (0.00830)	-0.0202 (0.0451)	-0.0686 (0.0663)	0.0686 (0.470)	-0.231** (0.109)	-0.0332*** (0.0114)
Dependent mean	0.051	0.310	0.337	0.824	1.298	0.051
Observations	13776	13776	13776	13776	8732	13776

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A11: Heterogeneity analysis for game day effects of Black quarterbacks

Heterogeneity characteristic:	(1) Black share	(2) Urban share	(3) Std RAI
Panel a: DMA daily hate crimes			
Below median characteristic areas			
<i>Post × Any Black QB</i>	0.00767 (0.0670)	0.0415 (0.0641)	0.0887** (0.0414)
<i>Post × Any Black QB × Win</i>	-0.0758 (0.0985)	-0.0524 (0.0961)	-0.101* (0.0581)
Differential for above median characteristic areas			
<i>Post × Any Black QB × Het</i>	0.115 (0.0767)	0.0585 (0.0739)	-0.0292 (0.0692)
<i>Post × Any Black QB × Win × Het</i>	-0.0469 (0.113)	-0.0692 (0.107)	0.00192 (0.0990)
Win below median ($\theta_2^b + \theta_4^b$)	-0.068	-0.011	-0.012
Loss above median ($\theta_2^b + \theta_6^b$)	0.123***	0.100***	0.060
Win above median ($\theta_2^b + \theta_4^b + \theta_6^b + \theta_8^b$)	0.000	-0.022	-0.040
Observations	44715	44715	36755
Panel b: Standardized IAT scores			
Below median characteristic areas			
<i>Post × Any Black QB</i>	0.0715** (0.0323)	0.0619** (0.0295)	0.0633* (0.0340)
<i>Post × Any Black QB × Win</i>	-0.0759 (0.0479)	-0.133*** (0.0450)	-0.0579 (0.0478)
Differential for above median characteristic areas			
<i>Post × Any Black QB × Het</i>	-0.0609 (0.0434)	-0.0384 (0.0438)	-0.0351 (0.0438)
<i>Post × Any Black QB × Win × Het</i>	0.00804 (0.0675)	0.117* (0.0666)	-0.0444 (0.0677)
Win below median ($\theta_2^b + \theta_4^b$)	-0.004	-0.071**	0.005
Loss above median ($\theta_2^b + \theta_6^b$)	0.011	0.024	0.028
Win above median ($\theta_2^b + \theta_4^b + \theta_6^b + \theta_8^b$)	-0.057	0.007	-0.074**
Observations	73129	73129	72914

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A12: Heterogeneity analysis for season effects of Black quarterbacks

Heterogeneity characteristic:	(1) Black share	(2) Urban share	(3) Std RAI
Panel a: DMA monthly hate crimes			
Below median characteristic areas			
<i>Main QB is Black $\times$ After season</i>	-0.00752 (0.00821)	-0.00851 (0.00575)	0.00364 (0.00504)
<i>Main QB is Black $\times$ After season $\times$ In championship</i>	0.00858 (0.01214)	-0.01001 (0.01430)	-0.02017* (0.01146)
Differential for above median characteristic areas			
<i>Main QB is Black $\times$ After season $\times$ Het</i>	0.01024 (0.00909)	0.01072 (0.00696)	-0.00946 (0.00847)
<i>Main QB is Black $\times$ After season $\times$ In championship $\times$ Het</i>	-0.04087*** (0.01343)	-0.00938 (0.01406)	0.00415 (0.01473)
Compared to white teams			
Below median – championship ($\zeta_A^r + \zeta_A^c$)	0.001	-0.019	-0.017
Above median – no playoffs ($\zeta_A^r + \zeta_A^c + XXXXX$)	0.003	0.002	-0.006
Above median – championship ($\zeta_A^r + \zeta_A^c + XXXXX$)	-0.030***	-0.017*	-0.022**
Observations	52044	52044	42972
Panel b: Google slur searches			
Below median characteristic areas			
<i>Main QB is Black $\times$ After season</i>	0.05984 (0.04644)	-0.01213 (0.08238)	0.08567*** (0.03305)
<i>Main QB is Black $\times$ After season $\times$ In championship</i>	-0.09767 (0.08998)	0.09383 (0.12793)	-0.20524*** (0.06072)
Differential for above median characteristic areas			
<i>Main QB is Black $\times$ After season $\times$ Het</i>	-0.00133 (0.05747)	0.08277 (0.08444)	-0.10268* (0.06004)
<i>Main QB is Black $\times$ After season $\times$ In championship $\times$ Het</i>	-0.06436 (0.11900)	-0.30035** (0.13048)	0.22239* (0.11633)
Compared to white teams			
Below median – championship ($\zeta_A^r + \zeta_A^c$)	-0.038	0.082	-0.120**
Above median – no playoffs ($\zeta_A^r + \zeta_A^c + XXXXX$)	0.059*	0.071**	-0.017
Above median – championship ($\zeta_A^r + \zeta_A^c + XXXXX$)	-0.104	-0.136***	0.000
Observations	15414	15414	15060
Panel c: White IAT scores			
Below median characteristic areas			
<i>Main QB is Black $\times$ After season</i>	0.01605 (0.01017)	-0.00402 (0.01106)	0.00049 (0.00658)
<i>Main QB is Black $\times$ After season $\times$ In championship</i>	-0.08126*** (0.02009)	-0.03742 (0.02296)	-0.04622*** (0.01546)
Differential for above median characteristic areas			
<i>Main QB is Black $\times$ After season $\times$ Het</i>	-0.02161* (0.01132)	0.00773 (0.01144)	0.00348 (0.01203)
<i>Main QB is Black $\times$ After season $\times$ In championship $\times$ Het</i>	0.06138*** (0.02194)	-0.00526 (0.02609)	0.01542 (0.02157)
Compared to white teams			
Below median – championship ($\zeta_A^r + \zeta_A^c$)	-0.065***	-0.041*	-0.046***
Above median – no playoffs ($\zeta_A^r + \zeta_A^c + XXXXX$)	-0.006	0.004	0.004
Above median – championship ($\zeta_A^r + \zeta_A^c + XXXXX$)	-0.025*	-0.039***	-0.027
Observations	3584160	3584160	3544543

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A13: Game day impacts of Black Head Coach

	(1) Hate crimes	(2) Submission removal	(3) IAT
White-white coach games:			
<i>Post</i>	0.0510 (0.0363)	0.0516** (0.0224)	0.0276 (0.0273)
<i>Post</i> × <i>Win</i>	-0.0257 (0.0294)	-0.0413*** (0.0149)	-0.0171 (0.0214)
Any Black coach games:			
<i>Post</i> × <i>Any Black coach</i>	-0.0481 (0.0398)	0.0166 (0.0179)	-0.0203 (0.0257)
<i>Post</i> × <i>Any Black coach</i> × <i>Win</i>	0.0706 (0.0553)	-0.0188 (0.0224)	-0.00448 (0.0398)
Observations	38865	2679	60535

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

Table A14: Season effects of local Black Head Coach by performance

	(1) Hate crime rate (monthly DMA)	(2) Google slur index (Standardized)	(3) Submission removals (Share)	(4) IAT scores (Standardized)
Not in playoffs				
<i>Head coach is Black</i> × <i>After season</i>	0.00425 (0.00449)	-0.00493 (0.0327)	-0.0000791 (0.00990)	0.00136 (0.00622)
In playoffs only				
<i>Head coach is Black</i> × <i>After season</i> × <i>In playoffs only</i>	0.00189 (0.00776)	0.0558 (0.0542)	-0.00457 (0.0213)	-0.00211 (0.0110)
In championships				
<i>Head coach is Black</i> × <i>After season</i> × <i>In championship</i>	-0.00538 (0.00771)	-0.0242 (0.0623)	-0.0287 (0.0232)	-0.00259 (0.0165)
Compared to white Head coach championship teams:	-0.001	-0.029	-0.029	-0.001
Dep. mean	0.0510	-0.146	0.103	0.168
Observations	54912	16266	3023	3584160
Team × seasons	704	635	252	618

Note: Standard errors are reported in parentheses, with the following significance indicators: * p<0.1, ** p<0.05 and *** p<0.01.

A2 Appendix Figures

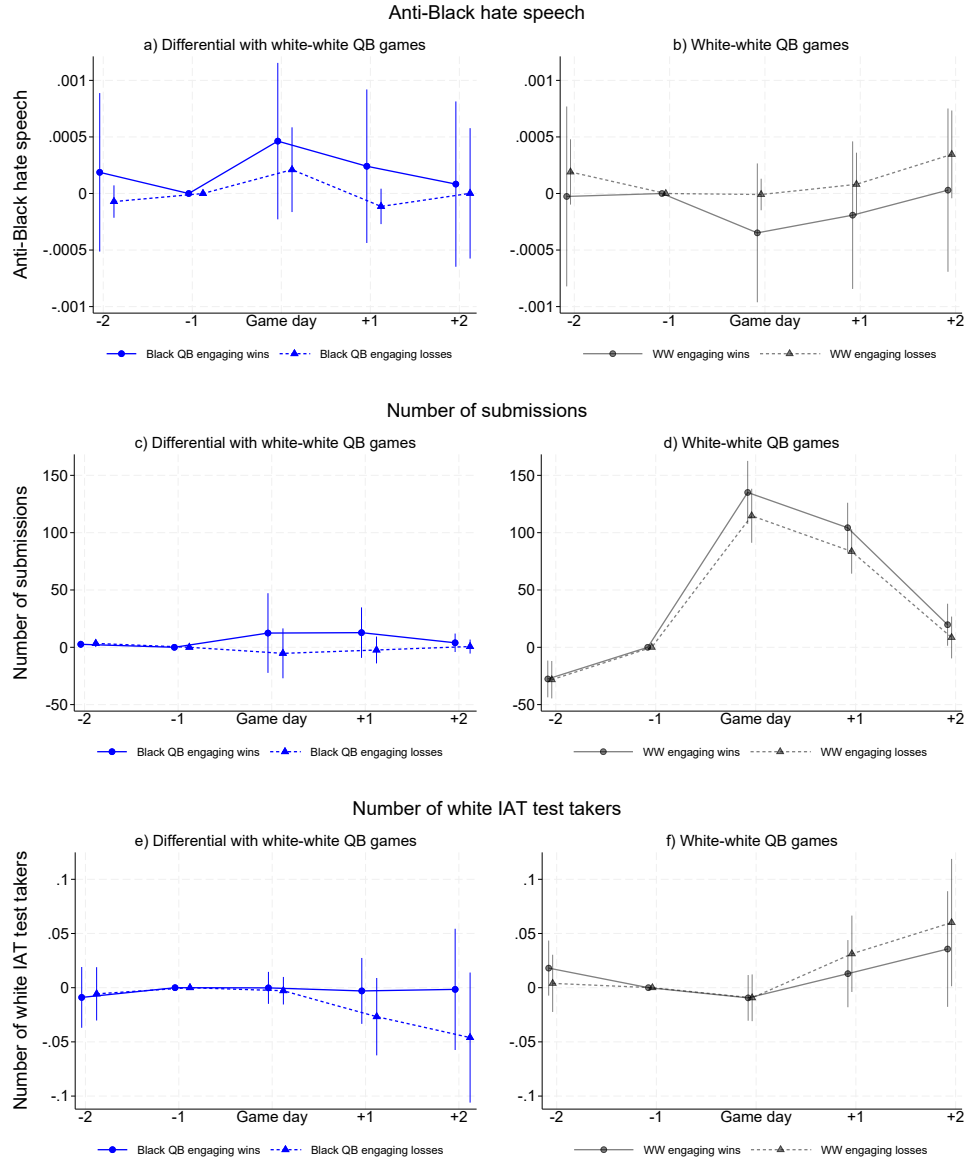


Figure A1: Game day effects of local Black quarterbacks

Figures on the left plot the differential effect of local Black quarterbacks for engaging games as estimated by equation 6. The ρ_2^b estimates are plotted for engaging losses and the $(\rho_2^b + \rho_4^b)$ estimates for engaging wins. Figures on the right plot the ρ_1^w estimates and the $(\rho_1^w + \rho_3^w)$ estimates showing the effects of engaging losses and wins around white-white quarterback games.

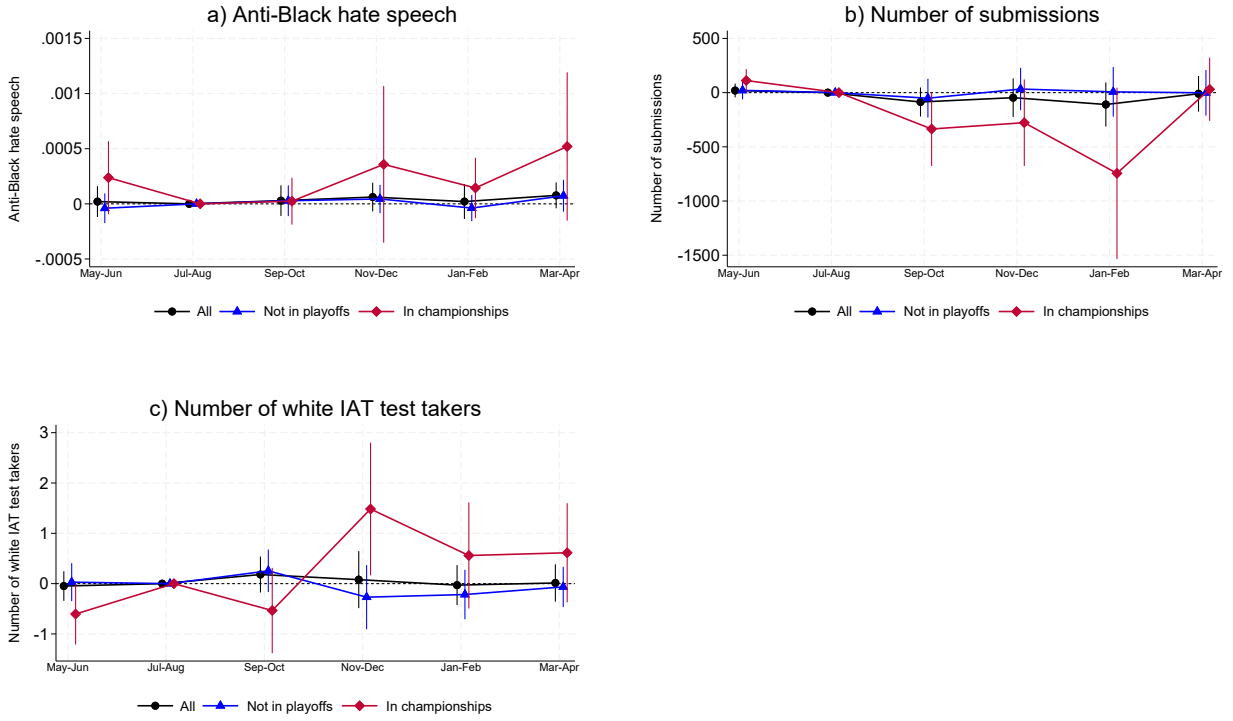


Figure A2: Long-run effects of local Black QBs

Notes: Figures plot the μ_T^r and μ_T^c from equation 7 in blue and red respectively. Aggregate estimates for all performance levels are plotted in black. These estimates control for *location* $\times$ *season*, *month* $\times$ *location*, *month* $\times$ *record*, and *month* $\times$ *season* fixed effects.

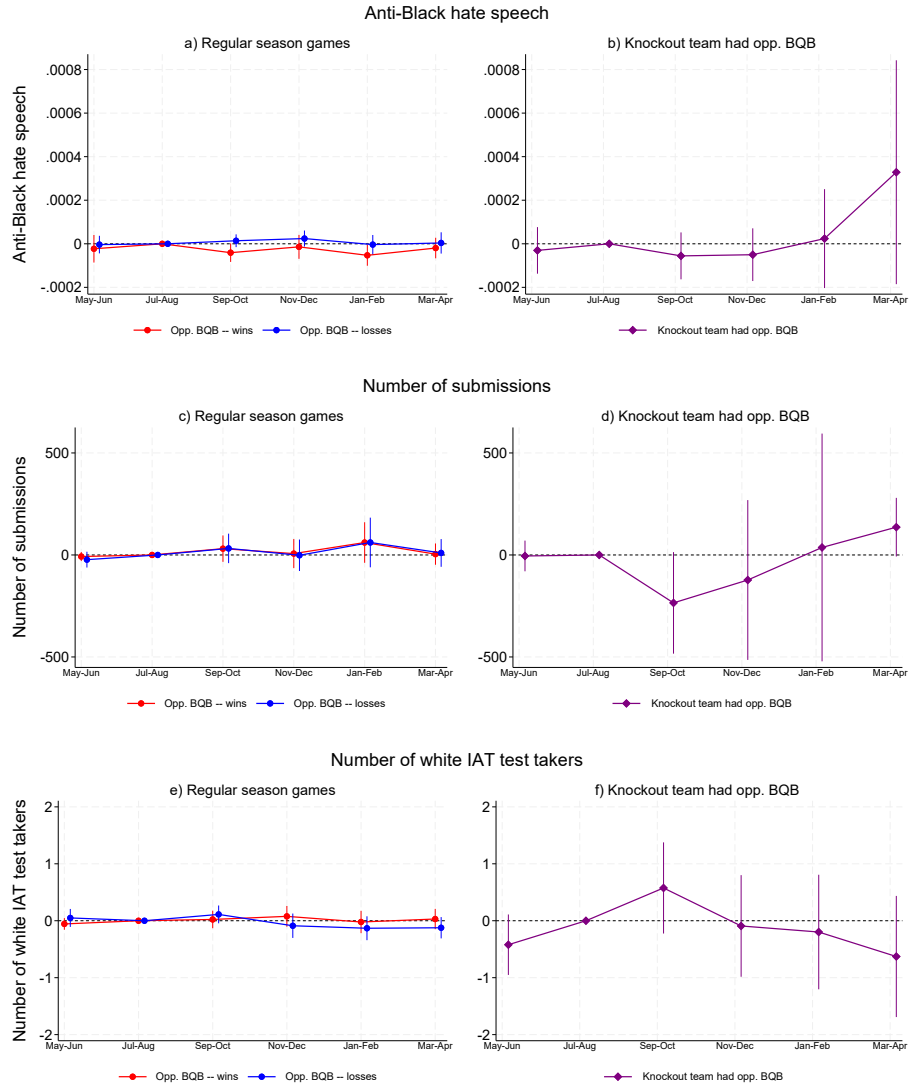


Figure A3: Long-run effects of opposition Black QBs

Notes: For all figures, the sample is limited to team-seasons where the main quarterback was white. Figures on the left plot the differential effect of an additional regular season game played against opposition Black quarterbacks. Figures on the right plot the differential effect of being knocked out of the playoffs by a team with an Black quarterback.

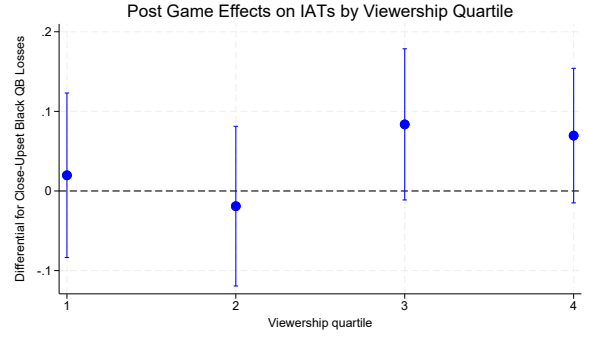
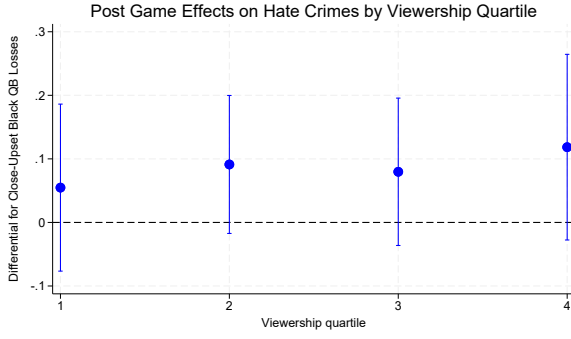


Figure A4: NEW quartile effects TO DISCUSS

Notes: Figures plot the μ_T^r and μ_T^c from equation 7 in blue and red respectively. Aggregate estimates for all performance levels are plotted in black. These estimates control for $location \times season$, $month \times location$, $month \times record$, and $month \times season$ fixed effects.

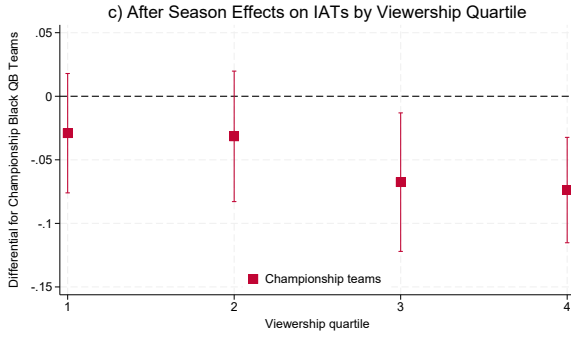
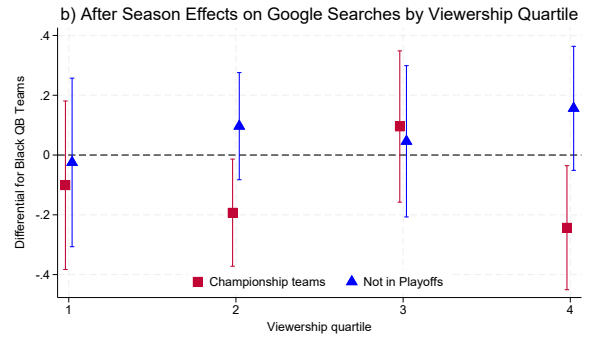
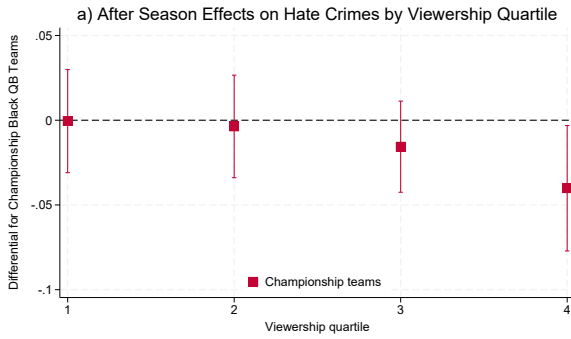


Figure A5: NEW quartile effects TO DISCUSS

Notes: Figures plot the μ_T^r and μ_T^c from equation 7 in blue and red respectively. Aggregate estimates for all performance levels are plotted in black. These estimates control for $location \times season$, $month \times location$, $month \times record$, and $month \times season$ fixed effects.

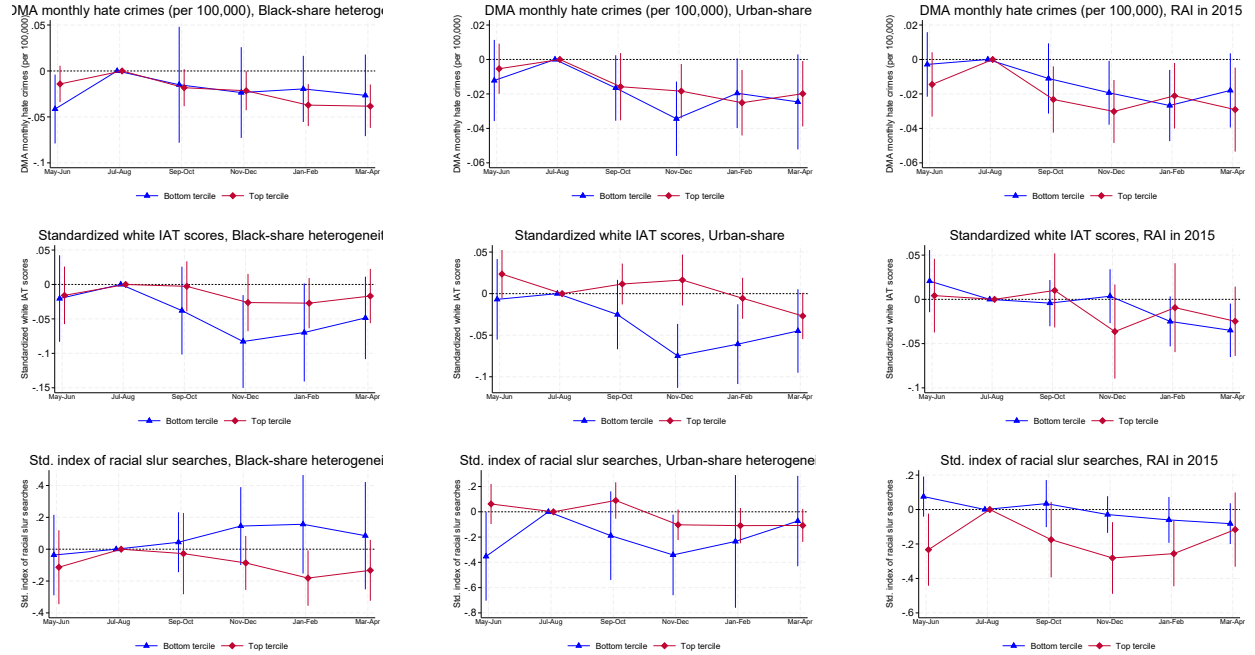


Figure A6: het fig

Notes: Figures plot the μ_T^r and μ_T^c from equation 7 in blue and red respectively. Aggregate estimates for all performance levels are plotted in black. These estimates control for $location \times season$, $month \times location$, $month \times record$, and $month \times season$ fixed effects.

A3 Quarterback style

Game-day: season: We use a conservative behavioral threshold. A quarterback-season is classified as mobility-integrated if mobility plays account for at least 10 percent of quarterback style plays and if the quarterback records at least 20 mobility plays. The share threshold ensures that mobility is a meaningful component of the quarterback's role, while the count threshold prevents small-sample classifications.